

The fine art of fine-tuning: A structured review of advanced LLM fine-tuning techniques

Samar Pratap^a, Alston Richard Aranha^a, Divyanshu Kumar^a, Gautam Malhotra^{a,*},
Anantharaman Palacode Narayana Iyer^b, Shylaja S.S.^a

^a Department of Computer Science, PES University, Bengaluru, 560085, Karnataka, India

^b JNRResearch, Bengaluru, 560085, Karnataka, India

ARTICLE INFO

Keywords:

Adapter
FFT
LLM
LoRA
MoE
PEFT
Quantization

ABSTRACT

Transformer-based models have consistently demonstrated superior accuracy compared to various traditional models across a range of downstream tasks. However, due to their large nature, training or fine-tuning them for specific tasks has heavy computational and memory demands. This causes the creation of specialized transformer-based models to be almost impossible in the generally present constrained scenarios. To tackle this issue and to make these large models more accessible, a plethora of techniques have been developed. In this study, we will be reviewing the types of techniques developed, their impacts and benefits concerning performance and resource usage along with the latest developments in the domain. We have broadly categorized these techniques into six key areas: Changes in Training Method, Changes in Adapter, Quantization, Parameter Selection, Mixture of Experts, and Application based methods. We collated the results of various techniques on common benchmarks and also evaluated their performance on different datasets and base models.

1. Introduction

With the emergence of LLMs (Large Language Models), the scene of NLP (Natural Language Processing) has dramatically changed and has made it possible to achieve performance never seen on most tasks. However, fine-tuning is called for on the application side to bridge such powerful models into practice on applications, adapting the pre-trained models to new tasks or domains more efficiently. While the research community continues to invent new solutions, a wide variety of fine-tuning methodologies defines their own advantages and difficulties. The aim of this survey paper is to explore and then classify these different methodologies, drawing insights from recent works and seminal works in this field.

Traditionally, LLM fine-tuning methods are categorized based on various criteria such as task-specific approaches, parametric efficiency, or architectural modifications. We further divide the fine-tuning methods for LLMs into six major classes: Application-Based Approaches, Changes in the Adapter, Changes in Training Method, Parameter Selection, Quantization methods, and Mixture of Experts. Such a structured classification not only brings clarity in understanding the landscape but also brings out innovative strategies that have been charting the course that the future of LLM fine-tuning is going to take. Naturally,

all discussions in the later sections of the paper are done in line with this classification.

Changes in Training Methods: The training process plays a pivotal role in determining the performance of all machine learning and deep learning systems. As discussed before, with the emergence of transformer-based models, many new challenges in the form of computational resource constraints and limited performance improvement have emerged. The traditional training methods are unable to effectively address these issues leading to the need for training methods specialized for these types of models which can perform well in the given resource constraints and at the required scale. In this category, we will be reviewing some of these proposed methods.

Parameter Selection: Since the computation cost of full fine-tuning is very high and fine-tuning all parameters may not necessarily benefit the fine-tuning process and may sometimes even impact it negatively, techniques have been developed to select the parameters to be fine tuned. Different techniques take into consideration, various sets of metrics to determine the parameters to be trained, leading to differing degrees of improvement or in some cases degradation with respect to a variety of metrics, some of these techniques shall be discussed in the later sections.

* Corresponding author.

E-mail address: malhotragautam3000@gmail.com (G. Malhotra).

Quantization Methods: One of the major challenges faced while fine-tuning LLM's is the limited amount of available memory. LLM's due to their large nature require a lot of memory and information like gradients and so required during the fine-tuning process further increasing the demand for memory required to fine-tune these models. One of the most effective methods introduced to tackle this problem is the quantization of the weights of the pre-trained models, called Q8BERT (Quantized 8-Bit BERT) (Zafir et al., 2019). This reduces the overall memory requirement of the models, enabling us to load and fine-tune very large models even in memory-constrained scenarios.

Changes to the Adapter: Some of the previous fine-tuning methods that changed the original model parameters or the internal structure of the original model faced difficulties during the training and deployment process. To solve these issues, the idea of only training a separate new module called adapter during the fine-tuning process, was introduced, named Parameter-Efficient Transfer Learning for NLP (Houlsby et al., 2019). Further works in the domain showed that the structure of these adapters affected the result significantly and that different applications had different optimal adapters in terms of size and structure. In this category this review delves into some of the proposed structures along with their benefits and use cases.

Mixture of Experts: Recently, in the field of language models, Mixture of Experts (MoE) has evolved as a resource efficient alternative. This has both advantages, a huge amount of information as possessed by large models and less memory requirement as that of small models. It does so by not loading all the parameters at once. The model is internally divided into multiple "experts" who know specific domains. During the prediction process, the input is routed to the suitable expert, leaving the other experts unactivated, thus, saving both memory and processing resources. The performance of these models has been comparable to the predecessors having a similar number of parameters as proven by various models like Mistral 7B (Jiang et al., 2023) developed by Mistral AI. This idea of the mixture of experts has also been adopted and explored in the field of fine-tuning, leading to the development of various techniques.

Application-Based Methods: Some of the explored techniques in the domain of fine-tuning are not generic and cannot be applied to all fine-tuning scenarios. In this category, we will be concentrating on these techniques which are specifically designed to optimize the fine-tuning process in certain scenarios where the general fine-tuning techniques cannot be applied or where their results are unable to meet the desired standards.

The framework helps to underline distinction across application-specific adaptations, architectural changes, training method modifications, MoE approaches, and parameter optimization techniques significantly impact the different strategies researchers might adopt to efficiently adapt LLMs to multiple tasks. This approach also helps to make a much clearer comparison between various methods as regards their respective benefits and limitations, showing the most appropriate fine-tuning strategy for the target needs. This review takes into account many research papers, including some pivotal ones, for instance, "Parameter-Efficient Fine-Tuning Methods for Pre-trained Language Models" by Xu et al. (2023a) and "Scaling Down to Scale Up" by Lialin et al. (2023), which give a panoramic view of the present status and future directions of fine-tuning LLMs. Classifying these methods and analyzing them bring out useful insight and guidance for researchers and practitioners to improve the performances of LLMs in many kinds of applications. In this regard, this paper provides a further structured overview of fine-tuning strategies that would go ahead to facilitate further advancements and innovations in the optimization strategies for large language models.

Our key contributions are as follows:

- We present a detailed review of developments in the domain of LLM fine-tuning.

- Our classification method provides more detailed insights into fine-tuning compared to existing classifications, offering an all-inclusive framework that includes various fine-tuning strategies.
- We classify and compare a large number of research papers, detailing the methodologies involved.
- Our review includes structured technical information about various methods and novel approaches, covering important areas such as performance on standard metrics, sizes, and memory usage. This information can help practitioners looking to make their models more efficient for specific use cases.

2. Preliminaries

2.1. Transformers

Previous sequence processing models (Elman, 1990; Graves and Graves, 2012; Cho, 2014; Sutskever, 2014) had the problem of forgetting past information in the sequence and taking a lot of time to process the sequence which limited their applicability. These issues were addressed by Vaswani (2017) with the introduction of the transformer architecture. This architecture solved the problem of forgetting past information with the help of the attention mechanism through which, only information relevant to the current context was taken, which reduced the amount of total information needed to be remembered at once. The problem of processing time was handled by parallel processing of the entire sequence.

In this new architecture, each token was converted into three vectors, namely query (Q), key (K) and, value (V) based on the token and its position in the sequence.

$$Q = XW_q + b_q, K = XW_k + b_k, V = XW_v + b_v \quad (1)$$

Where X is the word and positional embeddings of the tokens and W and b are the weights and biases used to calculate Q, K and, V. The attention for each token was then calculated by appropriately aggregating the information from all the tokens using these three vectors.

$$Attn(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (2)$$

Where d is the dimension of the embeddings used. The calculated attention contains the information held by the token in the given context and is used for the calculation of the output.

This breakthrough led to rapid innovation in the domain of sequential tasks, leading to the development of a series of transformer-based models with the capacity to learn information at an unprecedented scale. This has also brought in a new set of challenges as discussed earlier, many of which were addressed by the work discussed in the later sections.

2.2. Full fine-tuning

Initially, to adapt these large transformer-based models to specific downstream tasks, the approach followed was full fine-tuning. In this approach, the pre-trained model weights with a focus on generalization were taken as the initial weights, and the entire model, that is, all its layers, modules, and parameters were trained on the task-specific dataset, generally using techniques such as back-propagation and gradient descent (Tayaranian et al., 2022; Amari, 1993).

Although this method was effective in terms of the results obtained, the available computational resources were unable to scale with the growing size of the models and the amount of data required to train them. This was because the latest models had parameters on the order of billions and training all these parameters required a lot of memory and processing power, making the process of fine-tuning cost-inefficient. This led to the development of methods that trained only specific parts of the model (Zaken et al., 2021) during the fine-tuning process and methods such as LoRA (Hu et al., 2021) which introduced

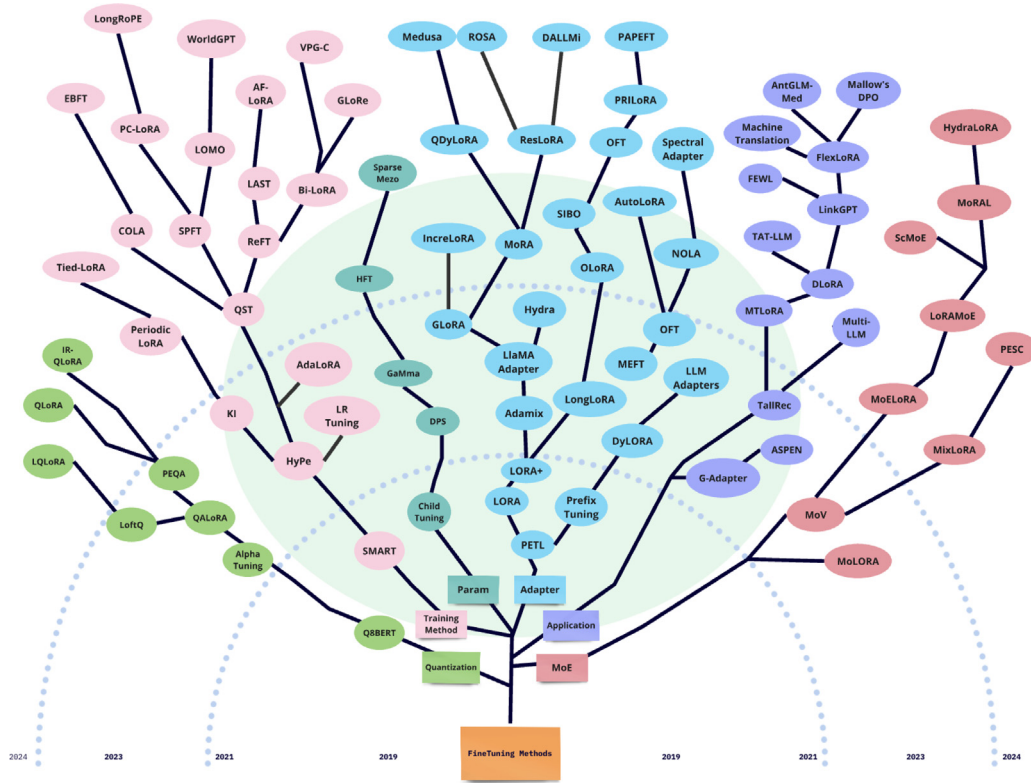


Fig. 1. The evolutionary development of fine-tuning methods in recent years. Models on the same branch have some common features. The distance from the root indicates the timeline of their release.

the idea of having separate modules called adapters to be trained during the fine-tuning process.

LoRA achieves this by introducing two trainable low-rank matrices A and B into the model structure. Instead of directly fine-tuning the weight matrix W , the input x is first projected into a lower-dimensional space using matrix A , then re-projected back to the original dimension using matrix B . The final result h is added to the original model output.

3. Fine-tuning methods

A tree representation has been drawn to demonstrate the evolution of fine-tuning methods over the years across different branches comprehensively. (see Fig. 1).

3.1. Changes in training method

This is a group of methods that try to change the training regime to achieve efficiency in fine-tuning. Examples may include novel optimization techniques, innovative data augmentation strategies, or hybrid training methods. Methods that deal with forward pass fine-tuning and gradient optimization have proven to be quite effective in this domain.

Fine-Tuning with only Forward Passes (Malladi et al., 2023) changes the method of determining the direction of the change in weight to perform the process with reduced use of memory and computation. Normally, when weights are updated using back-propagation, a lot of memory and processing power is consumed for storing and calculating the gradients needed for weight updation. In the proposed method, inference is made on training data with the current weights and with slightly altered weights, and the weights are updated on the basis of whether the results on the training data were better on the actual weights or the slightly altered weights. This has also been achieved by gradient optimization, where research like COLA (Diao et al., 2024) from Diao et al. proposes separation in the process of calculation of the hidden representations and parameters' gradients.

The authors also introduced Collaborative Adaptation along with the learning method and parameter merging to ensure a great decrease in the need for computational resources. There has been a lot of research on reducing the number of parameters and increasing the fine-tuning efficiency by employing Low rank adaptation methods.

In another approach named **PeriodicLoRA** (Meng et al., 2024), the authors tried to bridge the learning and performance gap between full fine-tuning and LoRA which is caused by its limitations of weight updates being low rank. It does so by training multiple low rank adapters and accumulating them to create a high rank adapter. The model is first trained with a low rank adapter and then, the low rank adapter is integrated with the backbone parameters of the model and the new model is trained with a new low rank adapter. Low rank adaptation is seen in **PC-LoRA** (Hwang et al., 2024). To facilitate parameter-efficient fine-tuning and model compression, a method of fading model weights to zero called PC-LoRA is introduced. It reduces the computing burden of large-scale models, by gradually lowering the pre-trained weights during the fine-tuning process until they are eliminated. Methods like **Tied-LoRA** (Renduchintala et al., 2023) involve using these adaptation models to reduce the number of parameters. Tied-LoRA is a new paradigm that improves the parameter efficiency of LoRA by utilizing weight tying and selective training. Original LoRA visualizes matrices Q , K , V as separate low-rank projections, whereas, TiedLoRA projects Q , K , V as a shared down projection matrix A and an up projection matrix B , reducing the number of parameters requiring training. By selectively updating in various combinations, task specific models with minimal training time can be obtained.

Optimizations involving projection matrices have been made using freezing, one such paper, **AFLoRA**, adaptive freezing of low rank adaptation in PEFT (Parameter Efficient Fine-tuning) of Large Models (Liu et al., 2024a) by Liu et al. introduces a novel parameter efficient fine-tuning approach via adaptive freezing of Low Rank Adaptation. Based on a self proposed new freezing score, projection matrices are iteratively frozen to reduce the computational resources required

and avoid overfitting. ALoRA has four major components: the down-projection linear layer, the up-projection linear layer, and two feature transform vectors positioned before and after LoRA block. Despite these improvements, overfitting can be an issue and to mitigate that, further research like **BiLoRA**, a Bi-level Optimization Framework for Overfitting-Resilient Low-Rank Adaptation of Large Pre-trained Models (Qiang et al., 2024). The work discusses that though LoRA vastly reduced the parameters to be trained compared to full fine-tuning methods, it tends to overfit, resulting in subpar performance on testing scenarios. To overcome this BiLoRA was introduced, which focuses on bi-level optimizations (BLO). It splits the singular vectors and values across two separate spheres of available training data, with different levels of BLO framework. This prevents the model to memorize results.

Parallely, techniques to reduce the memory footprint of models have been presented. **QST (Quantized Side Tuning)** Zhang et al. (2024a) introduces a specialized side fine tuning technique that does 4-bit quantization on the original model and uses a separate side network to avoid performing backpropagation through the LLM. By doing so, the memory required by the model and the intermediate activations during the fine-tuning process is saved, thus, making it more memory efficient. Another related research has been done by authors in **Low-rank Attention Side-Tuning** (Tang et al., 2024) for Parameter-Efficient Fine-Tuning where a side network is trained consisting only from low-rank attention modules. This approach of training makes the process highly parallel across various downstream tasks.

In the case of NER tasks, **Label Supervised LLaMA Finetuning** (Li et al., 2023a) proposes a new approach for tuning large language models, specifically LLaMA-2-7B, for discriminant label classification: Label Supervised LLaMA. Most instruction-tuned language models have suboptimal performance for both sequence and token classification tasks because they are trained for generating human-like responses rather than predicting exact labels. LS-LLaMA projects latent representations from LLMs into the space of labels and further refines them using Low-Rank Adaptation. This method outperforms significantly greater LLMs in text classification and even robust baselines such as BERT-Large. Moreover, removing the causal mask from the LLaMA decoder allows it to excel at NER tasks, reaching state-of-the-art results. Partially training models also reduce memory footprint, as shown in **ReFT (Representation Finetuning for Language Models)** (Wu et al., 2024a). The authors proposed a family of representational methods where the methods train the portions that influence only a small part of required model representations. One of the major contributions from ReFT includes LoReFT (Low-rank Linear Subspace ReFT) which works in a linear subspace of hidden representations. While another model from the family called DiReFT focuses on efficiency lowering the performance. Similarly, Lv et al. (2023) addresses the challenge of tuning the full parameters of LLMs with limited resources, by introducing a new optimizer, Low-Memory Optimization (LOMO), which fuses the gradient computation and the parameter update in one step to reduce memory usage. The concept behind LOMO is to update the parameter as and when the gradient is computed so that there is no need to store gradient tensor in memory.

Solutions to overfitting and resource optimization are also achieved through adaptive and hybrid training strategies. **ADALORA** (Zhang et al., 2023a) uses adaptive budget allocation through singular value decomposition for parameter-efficient fine-tuning. Another influential method is that of LARS, which makes its learning rates dynamic to further improve the model convergence in fine-tuning. Such strategies only turn out to be imperative training methodologies that are important for fine-tuning optimization. While another research **SMART** (Jiang et al., 2019) proposes a change in the aggressive training employed in language models for limited resources to prevent this overfitting and resource wastage. As a solution to this, the authors proposed smoothness-inducing regularization and Bregman proximal point optimization. This method of regularization was introduced and used in

Virtual adversarial training: a regularization method for supervised and semi-supervised learning for semi-supervised learning.

$$R_s(\theta) = \frac{1}{n} \sum_{i=1}^n \max_{\|\tilde{x}_i - x_i\|_p \leq \epsilon} \mathcal{L}_s(f(\tilde{x}_i; \theta), f(x_i; \theta)) \quad (3)$$

Another approach **HyPe** (Yuan et al., 2022) was proposed to handle the same problem but with a different way of solution. The authors not only add random noise to the hidden representations fed into each transformer layer in the forward process of computation but also, perturbing the hidden representations in transformers to ensure performance of each transformer layer.

Innovative approaches like self-play and ontological reasoning introduce new ways to refine training data and enhance domain-specific performance. **Self-Play Fine-Tuning** (Chen et al., 2024a) proposed a completely novel approach for fine tuning the language models, named SPIN where the training process is controlled by the model to be trained itself. The training data for each iteration is refined by the language model, which generates synthetic data. During the process of training, a two player scenario consisting of the main player and the opponent player is created. The training approach of 2 players aligns with the approach followed in a naive Generative Adversarial Networks (GAN). For this task, another paper **Selective Reflection-Tuning** (Li et al., 2024a) has been introduced. This paper proposes a new methodology, selective reflection tuning, to improve instruction tuning in LLMs. Under this approach, an LLM teacher is used to further refine existing data by advanced reflection and introspection in a teacher-student setup. This procedure enhances the data quality and aligns it with the capability of the student model so that efficient and high-performance instruction tuning without collecting new data would be possible.

Ontological reasoning has been employed in **Fine-Tuning Large Enterprise Language Models via Ontological Reasoning** (Baldazzi et al., 2023). This paper provides another way of fine-tuning Large Language Models through ontological reasoning induced from Enterprise Knowledge Graphs-prepend. Traditional ways of fine-tuning LLMs are often realized by general textual corpora, usually ending with limited performance on domain-specific tasks, as in finance or genomics. The paper describes an approach to making LLMs more specialized in enterprise domains through the integration of EKG-based ontological reasoning. As suggested, the proposed neuro-symbolic architecture is to exploit Datalog-driven reasoning for the synthesis of task-specific corpus functions by attacking tasks such as question answering or translation.

When training LLMs, improving the context handling and reasoning capabilities of LLMs has also been a focus, with methods extending context windows and refining reasoning steps.

Adapting LLMs for Efficient Context Processing through Soft Prompt Compression (Wang et al., 2024) proposed Soft Prompt Compression for cost-effective context processing. The technique integrates natural language summarization with soft prompts to encode them as the input for LLMs. The idea is about minimizing the computational overhead while retaining essential information for the downstream NLP tasks. It significantly showed improvements across a whole range of tasks like summarization, question answering, sentiment analysis and classification. Traditional transformer models struggle with long sequences due to the self attention mechanisms and quadratic complexity. **Longformer: The Long-Document Transformer** (Beltagy et al., 2020) introduced using a combination of local windowed attention and global attention, therefore improving the efficiency of the transformer for long sequences. **Big Bird: Transformers for Longer Sequences** (Zaheer et al., 2020) introduces a sparse attention mechanism that reduces this complexity to linear, making it more efficient for processing long sequences. It makes use of a random graph, each query attends to a fixed number of randomly chosen keys, rather than attending to all tokens in the sequence. BIGBIRD's attention is modeled as a directed graph, where nodes represent input tokens and edges represent attention interactions. Each token attends to a set of neighboring tokens

(local attention) and a small number of randomly selected tokens (random attention). **Parallel Context Windows for Large Language Models** (Ratner et al., 2022) splits long text into smaller chunks and reuses positional embeddings across them, allowing off-the-shelf LLMs to process longer sequences. It modifies the attention mechanism to ensure tokens within each window attend only to their own context, while task tokens can access all windows. **RoFormer** (Su et al., 2024) introduced a novel approach to the integration of position information in transformers, referred to as Rotary Position Embedding, RoPE. They manage to encode positions using a rotation matrix, thereby considering relative position dependencies in self-attention. RoPE is efficient, flexible, and decays inter-token dependencies over increasing distances. **LongRoPE** (Ding et al., 2024) extends LLM context windows beyond 2M tokens through nonuniform positional interpolation. Evolutionary search optimizes RoPE rescale factors by exploiting dimension and position non-uniformities, which enables no-fine-tuning 8x extensions. Then, the progressive strategy reaches 2048k contexts by extending to 256k, briefly fine-tuning, and further extending to 2048k. It improves over existing methods, recovering short-context performance. Tested on LLaMA2 and Mistral across a span of tasks.

In order to refine reasoning, **GLoRe** (Havrilla et al., 2024) has been introduced. This paper presents Stepwise Outcome-Based Reward Models to guide LLM reasoning on when, where, and how to refine solutions. It ensures, compared to the more traditional environment-based Reward Models, SORMs better predict the correctness of the intermediate steps in reasoning when trained on the synthetic data. The models developed by the authors are global refinement to correct the solution as a whole and local refinement to correct certain errors identified by SORMs. Preserving in-context learning ability is a major part of context handling. Research in this aspect has been done in **Preserving In-Context Learning Ability in Large Language Model Fine-tuning** (Wang et al., 2022a). The paper addressed how LLMs lose their in-context learning abilities during fine-tuning, caused by format specialization. Their solution is very straightforward: first, tune a soft prompt, keeping the model frozen; then, fine-tune the model attached with the prompt, called Prompt Tuning with Model Tuning or ProMoT. This methodology allows for the preservation of in-context capabilities by shifting the burden of format learning onto the prompt.

Further innovations address specific challenges like meta-training, class imbalance, hallucinations, positional bias and various issues pertaining multimodal architectures. **MAML-en-LLM** (Sinha et al., 2024) presents MAML-en-LLM, a new meta-training approach to Large Language Models that aims to enhance in-context learning without fine-tuning. While traditional meta-training methods, such as MetaICL and MetaICT, have trained LLMs on a wide variety of tasks, most often failing to attain truly generalizable parameters, MAML-en-LLM adapts models of LLMs in a two-step optimization process. Hallucination is a big issue in Large Language Models and is a topic heavily researched on. Some of the noteworthy works are **Knowledge Injection to Counter Large Language Model (LLM) Hallucination** (Martino et al., 2023). Large language models often output false information, which is an enterprise problem when reliable and accurate content is needed. This paper proposes Knowledge Injection as a way to remedy the problem of LLM hallucinations by integrating, at the very center, data from the Knowledge Graph inside prompts. Compared to generating responses to online customer reviews for retail locations, KI showed an increased accuracy of the generated text. Fine-tuned bloom-560m with KI performed better compared to a much larger non-fine-tuned model of text-davinci-003 from OpenAI, proving that KI has huge potential to further improve LLM performance while cutting down on cost. Another such approach, presented by Ji et al. (2023) designed an iterative self-reflection method to reduce AI-generated content hallucinations. The methodology enhances factuality, consistency and entailment of the answers considering that LLMs can interact and perform multitasking for gradual improvements. It refers to three loops: Factual Knowledge Acquiring, Knowledge-Consistent Answering, and Question-Entailment

Answering. Thus, it goes into an answering loop to evaluate entailment to ensure the questions are in agreement with the answers. Blum (2023) points out the “perplexity curse” in LLMs: they do a pretty nice job at minimizing perplexity but have a hard time extracting information from fine-tuning documents. Their conclusion is that the cause is auto-regressive training and that there is a need to introduce datasets to study QA performance relative to answer position. It can improve information extraction using regularization techniques, such as de-noising auto-regressive loss. These results have important implications for how to extract knowledge from LLMs and how to trade off RAG versus fine-tuning for domain adaptation.

WorldGPT (Ge et al., 2024) introduces WorldGPT, a generalist world model derived from multimodal large language models, and learns the world’s dynamics across domains and modalities from millions of videos. That is, the cognitive architecture that includes memory offloading, knowledge retrieval, and context reflecting is fused into the model to enhance performance in out-of-distribution scenarios. WorldGPT can thus precisely model state transitions in complicated situations and become a potential world simulator for multimodal agents. The authors introduce WorldNet—a general benchmark suite for multimodal state transition prediction and explore dream tuning for training data synthesis. Among similar works, **Fine-tuning Multimodal LLMs to Follow Zero-shot Demonstrative Instructions** (Li et al., 2023b) proposes a Complete Visual Prompt Generator module, VPG-C, to enhance the zero-shot demonstrative instructions in MLLMs. Traditional VPGs are concerned only with primary visual content due to training on image-caption pairs and miss important details. These missing information and filling are inferred with the help of VPG-C. It is trained with a synthetic discriminative training strategy. Thus, VPG-C has proven to show high improvement in zero-shot metrics with respect to baseline and therefore has greater comprehension for complex multimodal instructions.

Finally, benchmarking and tuning frameworks provide structured approaches to optimizing learning rates and fine-tuning sparse models and are important tools in enhancing overall performance. **Rethinking Learning Rate Tuning in the Era of Large Language Models** (Jin et al., 2023) introduces LRBench++, a framework for benchmarking and tuning learning rate policies for DNNs and LLMs. It analyzes the differences between DNN training and LLM fine-tuning, adds LLM support to LRBench, and includes new components like an LR monitor, policy database, and tuning optimizations. LRBench++ supports various tasks and tuning methods, aiming to be an open platform for improving learning rate tuning across model types. LRBench++ enhances LRBench for learning rate (LR) tuning in DNNs and LLMs. It offers 3 tuning methods: single policy selection, multipolicy composition, and cost-effective tuning, aiming to optimize accuracy or minimize costs under given constraints. Less resource intensive approaches include **EBFT** (Effective and Block-Wise Fine-Tuning for Sparse LLMs) (Guo et al., 2024) is a novel approach for fine-tuning of Sparse LLMs based on reducing reconstruction error. It reduces resource requirements by needing only a small dataset, on which block-wise optimizations are individually applied, having a cap on number of iterations. This reduces dependence on heuristics or approximation methods.

3.2. Parameter selection

Methods in this class are designed to efficiently select and tune parameters in order to optimize model performance, while balancing memory efficiency and generality. Approaches include freezing most model parameters and selectively updating a subset to overcome problems such as catastrophic forgetting. For instance, HFT (Hui et al., 2024) periodically resets partial parameters to recover initial knowledge in LLMs, acting as a regularization mechanism and maintaining knowledge parity across sequential tasks.

Initially, the techniques in this method either had a fixed way for selecting parameters or randomly selected parameters. One such

work is **Prefix-Tuning** (Li and Liang, 2021) which optimizes task-specific vectors while keeping language model parameters frozen, further, underline the strategic importance of parameter selection in fine-tuning models. All these methods underline that intelligent handling of parameters is one of the crucial prerequisites for efficient model optimization.

Later works like **Fine-Tuning Pre-Trained Language Models Effectively by Optimizing Subnetworks Adaptively** (Zhang et al., 2022a) where the authors described how choosing subnetworks randomly ignore the importance of parameters, addressed the issue of random selection. They proposed a new algorithm named as Dynamic Parameter Selection (DPS) where a stage-specific subnetwork is formed. The algorithm has 2 versions for full network update process and stochastic parameter update process respectively.

Similarly, **Child Tuning** (Xu et al., 2021) updates only a subset of parameters of pre-trained models by masking out the gradients of the non-child network during the backward process. The authors proposed 2 different approaches to handle over-fitting and improved generalization in cases of both task-free and task-driven cases. **GamMa** (Gui et al., 2023) relies on gradient activation mapping to identify and modify class-specific parameters, allowing it to provide better selective adjustments in parameters.

Sparse MeZO (Liu et al., 2024b) proposes a memory-efficient zeroth-order optimization approach that requires only forward passes during training and therefore is more accessible to many users who have limited computational resources. Sparse MeZO applies zeroth-order optimization only to carefully chosen parameters; as a result, memory usage is reduced, and a simple, decent enough scheme for selecting parameters suffice to ensure major performance gains. In addition, a memory-optimized implementation of sparse masking allows fine-tuning of large models like LLaMA-30b on a single GPU. Sparse MeZO gains performance and accelerates the traditional MeZO protocol.

3.3. Quantization method

A list of methods has been applied by various authors to reduce computational complexity and reduce the memory usage of LLMs through quantization. Initial research in this domain involved trying different methods to quantize the weights, like **AlphaTuning** (Kwon et al., 2022) which introduces a special type of quantization in which binary-coding quantization is applied on the weights of the model. This factorizes the model weights into binary parameters and a separate set of scaling parameters. During the training process, the value of the binary quantized weights are frozen and only the set of scaling factors is fine-tuned. This method has given great results comparable to LoRA and full fine-tuning, both of which do not compress the model.

One of the most important works in this domain, which is considered a benchmark by many is **QLoRA** (Dettmers et al., 2024). It is an efficient Fine-Tuning of Quantized LLMs. It took forward the idea of quantizing the weights by introducing double quantization. In this method, the quantization constants used to quantize the weights of the model are also quantized to save the space needed to store this constant. In addition to double quantization, the work also introduces 4-bit NormalFloat, which was a new data type used to store quantized weights and paged optimizations to reduce memory spikes that occur while checking the model. Additionally, **QDyLoRA** (Rajabzadeh et al., 2024), inspired by the works of QLoRA and DyLoRA, aims to combine their benefits, that is, a high degree of quantization and dynamic determination of the optimal LoRA rank. The experiments performed by the authors show that the proposed method consistently outperforms both QLoRA and DyLoRA, but still falls short of full fine-tuning.

A further improvement in the performance of the proposed method can be done by making the degree of quantization dynamic.

Following the initial research, Elastic quantization methods like **Accurate LoRA-Fine-tuning Quantization of LLMs via Information Retention** (Qin et al., 2024) brought the idea to change the quantization

process dynamically in accordance with the downstream performance of the model. Their Quantization-Aware Training (QAT) methods have shown a significant overhead improvement in the models without affecting their accuracy. They insist that it is quantization that makes LLMs more deployable and efficient, finally seeing broad applications in many diverse environments. One representative example is **QA-LoRA** (Xu et al., 2023b), which increases the degree of freedom in quantization and refines the fine-tuning process. It tackles the issue that to merge adapter weights and quantized model weights, all the weights need to be converted to Floating Point 16 which can then be quantized again after merging for memory efficient deployment. This method has a disadvantage of causing loss of accuracy during the conversions. QA-LoRA solves this issue by using a new type of quantization method along with a new data-type INT 4 which enables the merging of the weights in the quantized form itself.

With respect to PEFT methods alongwith keeping a check on memory consumption, **MEFT** (Memory-Efficient Fine-Tuning) (Kim et al., 2024a) proposed Parameter-Efficient and Quantization-aware Adaptation (PEQA) proposed a 4-bit quantized model through PEQA, where the model size is decreased to one fourth of its size when compared to LoRA (QV4).

LoftQ (Li et al., 2023c) introduced a quantization framework that helps in quantizing the model and finding a suitable low rank initialization at the same time to preserve the information from the original model by usage of low-rank adapters. Similar to LoftQ, **LQ-LoRA** (Guo et al., 2023) overcomes the bottlenecks associated with the sheer size of pre-trained LLMs by using a low-rank plus quantized matrix decomposition approach. This method is based on the normal float quantization scheme. It works by breaking down each pre-trained matrix into a low-rank matrix and a quantized matrix using a straightforward factorization algorithm. Only the low-rank component is adjusted during fine-tuning. By weighting the reconstruction objective with a diagonal approximation of the Fisher information matrix, they also present a data-aware version of the method.

3.4. Changes in the adapter

Initially, research regarding adapters in fine-tuning mainly revolved around just trying different adapters at different positions. One of such works was **LLM-Adapters** (Hu et al., 2023) which considered in its study, Prefix-tuning, LoRA, Series adapter, and Parallel adapter. The optimal placements found for the series adapter were after the MLP layers, the parallel adapter was in parallel with the MLP layers and LoRA is after both the Attention layers and MLP layers simultaneously. Shortly later, **LlaMA adapter** (Zhang et al., 2023b) was proposed which focused on faster fine-tuning of LlaMA models with the usage of zero initialized attention mechanism. The authors also performed a performance comparison between the Alpaca model and a LlaMA adapter based model. Another research **Non-intrusive adaptation** (Wang et al., 2023) proposes flexible partial input adaptation by transforming only selected embedding to enable minimal interference across tasks. Later on, the focus shifted towards techniques that modify the architecture of the adapter. These methods alternate the structure of the adapter in the hope of finding a better configuration to improve model performance and adaptability.

As shown in Table 1, even though FFT has given the best performance across all tasks, it is observed that LoRA and its variants consistently give performance comparable to FFT across various tasks. More specifically, AsyLoRA and MoRA show promising performance across multiple domains across instruction tuning and mathematical reasoning, with MoRA producing notable results even in continual pretraining because of its ability to dynamically adapt to new domains. It is also observed that even after significantly increasing the rank of the adapters, the performance in Instruction Tuning across methods remains largely unchanged.

Table 1

Performance of FFT, LoRA, LoRA variants on instruction tuning, mathematical reasoning and continual pre-training tasks.

Method	Rank	Instruction tuning		Mathematical reasoning		Continual pretraining	
		MMLU 0	MMLU 5	GSM8K	MATH	BioMed.	Finance
FFT	–	50.6	51.3	66.6	20.1	56.4	69.6
LoRA	8	50.2	51.5	64.6	15.1	52.3	64.0
LoRA+	8	49.2	51.1	64.1	15.8	52.2	64.9
ReLoRA	8	49.3	50.2	61.5	14.5	46.3	61.0
AsyLoRA	8	50.3	52.2	64.5	15.0	52.5	63.5
DoRA	8	50.2	51.5	64.5	14.6	52.5	63.9
MoRA	8	49.7	51.5	64.2	15.4	53.3	67.1
LoRA	256	49.7	50.8	67.9	19.9	54.1	67.3
LoRA+	256	49.2	51.3	68.2	17.1	54.2	66.7
ReLoRA	256	–	–	64.0	18.1	52.9	57.9
AsyLoRA	256	50.1	52.0	66.9	19.3	54.1	66.9
DoRA	256	49.6	51.1	67.4	19.5	54.2	66.0
MoRA	256	49.9	51.4	67.9	19.2	55.4	68.7

LoRA was a breakthrough in the field of fine-tuning, substantially reducing memory footprint and ameliorating resource efficiency. Many researchers build upon LoRA, looking for optimizations or modifications that could further better its performance. **MoRA** (Jiang et al., 2024) is innovating in this space by employing a square matrix instead of low-rank matrices in LoRA to achieve high-rank updating for parameter efficient fine-tuning. It was realized to improve task-specific performance further by using Lightweight Fine-Tuning methods, which include modifying the adapter settings. All techniques show the flexibility of adapter-based modifications toward adaptability and efficiency in this model. **Delta LoRA** (Zi et al., 2023) Delta LoRA improves upon LoRA by propagating updates to pre-trained weights(W). To avoid storing first- and second-order momentum in the optimizer, it employs an innovative method using the delta of the product of two low-rank matrices. This approach learns more parameters than LoRA. Additionally, Delta LoRA eliminates the Dropout layer before low-rank matrices A and B, aligning the gradients for W and AB, which were previously misaligned due to dropout. **LoRA+** (Hayou et al., 2024) argues that the use of the same learning rate by matrices A and B in the original LoRA lead to reduced performance and poor feature learning, especially in cases when model width is large. Hence, LoRA+ is defined as LoRA with different learning rates for matrices A and B in a prior-defined fixed ratio. **OLoRA** (Büyükkayüz, 2024) accelerates the convergence process during the fine-tuning of the LLM through orthogonal initialization of the weight matrix through QR decomposition. The results of the experimentation done by the authors show that OLoRA significantly accelerates the convergence process and sometimes gives better performance compared with standard LoRA.

Many authors also toiled around introducing residuals to the original architecture. **ResLoRA** (Shi et al., 2024) proposed modified ResLoRA blocks inspired by ResNet, based on which, residual paths are added by usage of 3 types of blocks named: input-shortcut, block-shortcut, and middle-shortcut blocks. For the merging process of these shortcuts as LoRA blocks to merge into the linear layer eventually, 2 different approaches, based on Input and based on weights of ResLoRA blocks are proposed. Another recent approach for enhancing PEFT namely **SIBO** (Wen et al., 2024) (Simple Booster for Parameter-Efficient Fine-Tuning) by inserting an initial residual to each layer of the model. This prevents over-smoothing as is noted in PEFT. The authors have used the method in both Adapter and LoRA techniques and noted improved performance.

Some of the works were also based on finding the optimal LoRA ranks. **DyLoRA** (Valipour et al., 2022) tries to address the issue that different applications have a different optimal rank for the adapter and the only way to find this rank is to perform an exhaustive search. This is done by training the adapter within a range of ranks instead of a single rank by dynamically adjusting based on the learned representation during training. The authors of the paper **LongLoRA** (Chen et al., 2023) provided an efficient method to evade the usage of basic self-attention

which leads to computation savings. They made use sparse attention methods over self-attention blocks to extend the context of models such as Llama2. Their method of splitting the context into several groups also succeeded in reducing the number of trainable parameters to a great extent. **AutoLoRA** (Zhang et al., 2024b) follows automatic tuning of matrix ranks in LoRA. The proposed method ensures that an optimal rank is chosen for each update matrix. In this approach, those rank one matrices are removed, where its corresponding selection variable sums close to zero. Another approach to handle the lower bound in form of rank one matrices in LoRA has been proposed in **NOLA** (Koohepayegani et al., 2024). The proposed approach re-parameterizes the low-rank matrices using linear combinations of matrices which are randomly generated.

Some approaches took into consideration optimizing use of data per iteration and removal of insignificant ranks during the process. **PriLoRA** (Benedek and Wolf, 2024) focuses on improving PEFT by allocating increasing low-rank values across layers of the transformers linearly. This ensures that lower layers function well with lower ranks, thus making the approach parameter efficient. Additionally, selective pruning is done based on importance to focus on critical components thus ensuring a gain in performance. **ALoRA** (Liu et al., 2024c), on the other hand, dynamically adjust the adapters so that it can effectively allocate resources in terms of fine-tuning requirements by introducing AB-LoRA to calculate the importance score for each of the LoRA ranks. This enables the pruning of the LoRA ranks which did not contribute much to the fine-tuning process or negatively impact it. The pruned LoRA budget can further be allocated to the more important Transformer modules. While ALoRA focused on resources allocation, **Make Pre-trained Model Reversible** (Liao et al., 2024) proposed to reduce the activation memory by applying a reversible model, so the intermediate activations are not necessary to be stored as cache data and can easily be recomputed in the next iteration.

Spectral Adapter (Zhang and Pilanci, 2024) is built on using the spectrum information of pre-trained weight matrices, incorporating two methods of spectrum adaptation: additive tuning and orthogonal rotation of the top singular vectors. Both methods involve fine-tuning the top spectral space after performing Singular Value Decomposition (SVD) on pre-trained weights.

Certain approaches also kept into consideration the need for a good generalization performance. **Hydra** (Kim et al., 2024b) uses adapters over the pre-trained model both in a sequential and parallel manner. To ensure a better generalization across various desired tasks, a linear combination of the existing features is taken into consideration. **One-For-All** (Chavan et al., 2023) from Arnav et al. proposes GLoRA where standalone structures are used in each layer of transformers to learn individual adapters, respectively. **AdaMix** (Wang et al., 2022b) is a proposed general PEFT method where any of the PEFT methods can be used in each of the transformer layers, thus allowing a mixture of different adapters. The usage of multiple adapters encourages the pre-trained

model to gain multiple perspectives for a single task. Another study, **RoSA** (Nikdan et al., 2024) introduced a novel parameter-efficient fine-tuning approach based on robust principal component analysis. This method decomposes updates into two components, namely low-rank and sparse components.

Numerous researchers point out the presence of an inherent positional bias in LLMs, which stems from flattening operations. This bias dictates that the position where the correct information is stored in the ordered input can influence performance. Zhang et al. (2024c) resolve this issue by presenting **Position-Aware Parameter Efficient Fine-Tuning (PAPEFT)**. This approach combines a position-aware adapter module with data augmentation techniques to ensure equitable distribution of attention across input context.

Another recently proposed approach, **Parameter-Efficient Orthogonal Fine-Tuning Via Butterfly Factorization** (Liu et al., 2023) makes use of orthogonal fine-tuning. **Orthogonal Fine-Tuning** (Qiu et al., 2023) is a technique where relational information between neurons is preserved thus maintaining the structured nature of weight matrices. The method proposed by the authors works in the reduction of the number of trainable parameters by applying the Cooley–Tukey fast Fourier transform algorithm as butterfly structures.

Cai et al. (2024b) introduced **MEDUSA**, an efficient method for LLM inference using multiple decoding heads to predict tokens in parallel. Traditional autoregressive decoding faces challenges with high memory usage and sequential constraints. MEDUSA addresses these using a tree-based attention mechanism to generate and check token sequences in parallel. It has two variants: MEDUSA-1 for speeding up inference on a frozen backbone and MEDUSA-2, which fine-tunes extra heads for better speed and accuracy. Experiments exhibit important inference acceleration without sacrificing quality in generation. Similarly, **IncreLoRA** (Zhang et al., 2023c) introduced an incremental parameter allocation method that allows trainable parameters to be added dynamically during training as per module importance scores.

3.5. Mixture of experts

MoE-based methods require a pool of expert models that dynamically choose the most relevant expert to a task at hand during inference. As an example, **MoELoRA** (Luo et al., 2024), **MIXLORA** (Li et al., 2024b), and **LoRAMoE** (Dou et al., 2024) inject specialized LoRA-based experts into pre-trained models so that fine-tuning efficiency pertaining to distinct features learned via contrastive learning can be improved. On the other hand, Pushing Mixture of Experts to the Limit proposes extremely parameter-efficient MoE configurations combining lightweight experts for instruction efficient tuning. These techniques distribute the computational workload across devices by optimizing sparsely gated MoE models, and shortcut-connected expert parallelism. Innovations like **Chain of LORA** (Xia et al., 2024) for residual learning in language models and **HydraLoRA** (Tian et al., 2024) illustrate MoE's central role in balancing model capacity and computational efficiency. While, **Pushing Mixture of Experts to the Limit** (Zadouri et al., 2023) proposes usage of lightweight experts over the pre-trained model to adapt the pre-trained Transformer layers in the fine-tuning time, Wu et al. (2024b) uses Parameter-efficient sparsity crafting to create multiple MoE based models, to achieve greater performance. It also applied QLoRA as a technique of parameter updation of unimportant networks. These techniques have been further improved for the medical domain in **MOELoRA** (Liu et al., 2024d), which combines mixture-of-expert (MOE) for multi-task learning with low-rank adaptation (LoRA) for parameter-efficient fine-tuning. MOELoRA integrates MOE with LoRA via multiple experts, each comprising low-rank matrices. This design retains the small size of trainable parameters while retaining the diversity in medical tasks handling abilities of the model. A task-driven gate function controls dynamically the contribution of each expert, modulating model parameters towards specific tasks. Experiments on the multi-task medical dataset have shown that MOELoRA offers far

better performance compared to existing methods for efficient fine-tuning of parameters. MoE is further accelerated in approaches like **ScMoE** (Cai et al., 2024a) (Shortcut-connected Expert Parallelism for Accelerating Mixture-of-Experts). It proposes a shortcut-connected MoE architecture with overlapping parallel procedure. The novelty of this research is the dynamic scheduling of operators based on real-time performance metrics.

3.6. Application based

These are specific in nature and as such, optimize LLMs for certain kinds of applications, which could be in the form of sentiment analysis, machine translation, or question answering. For example, **MTLoRA** (Agiza et al., 2024), which suggests that in multi-task scenarios, instead of using a large adapter to store all the fine-tuned information, we use specialized multiple low rank adapters for each tasks. This provides multiple benefits like avoiding the heavy cost of training a huge adapter, having adapters specialized for a specific task and disentanglement between the information of various tasks which prevent the information regarding one task from interfering with the results of another task. Graphs play a central role in fields of Social Network Analysis, Search Engine Algorithms, dependency mapping, flow planning problems and many more. **G-Adapter** (Gui et al., 2024), which introduces graph structure as an inductive bias to guide the fine-tuning process. Not only, the proposed G-Adapter ensure performance for graph based tasks with the modified adapter but also prevents over-training of language models with the help of Bregman proximal point optimization. **LinkGPT** (He et al., 2024) : LinkGPT is trained on the task of link prediction in a TAG (Text Attributed Graph), given the graph structure and node attributes. It employs the use of dual encoding, first for the node with its surrounding neighbors, through contrastive graph text pre training and then pairwise encodings, through LPFormer for pairwise relationships between nodes. This work utilized a two stage instruction tuning approach for training on link prediction and neighbor prediction tasks. In the initial stage, the LLM is kept frozen, and only the encoding-related modules, including the pairwise encoder and two alignment projectors, are tuned. In the latter stage, the encoding-related modules are frozen and LLM is trained through LoRA. These approaches highlight the importance of customizing models to meet the unique requirements of different applications, maximizing their utility and performance.

Then there are certain techniques which focus solely on addressing data security concerns and improving transparency. **DLORA** (Gao and Zhang, 2024) enables scalable parameter-efficient fine-tuning operations to be performed collaboratively between the cloud and user devices, focusing on privacy. Such methods are driven by the need to tailor a model for the requirements of specific applications, maximizing their utility. Bai et al. (2024) explained how training any kind of Machine Learning model on enormous data store with a single centralized server raises data security and privacy concerns. To account for such complexities federated learning is promoted, this paper focused on utilizing the same concept for the similar concerns in the domain of a LLM. Concluding, their work introduced FlexLoRA, an aggregation framework for the fine-tuning of LLMs along with dynamic corrections in local LoRA ranks for individual clients. FlexLoRA also takes into consideration that resources with every client are efficiently utilized by abating the “bucket effect”. FlexLoRA also maintains a balance between the ability to generalize on unseen data and task specific performance by changing local ranks with respect to net available resources per client node. The global weights formed by averaging the local ranks, undergo single value decomposition and redistributed to clients for preserving information.

Then there are also cases when enough data, textual information or resources are not available for a LLM to be fine-tuned by a conventional course of action. This is where methodologies such as **Generalizable**

and Stable Fine-Tuning of Pretrained Language Models on Low-Resource Texts (Somayajula et al., 2024) enters the picture. This paper proposes a novel regularization method based on attention-guided weight mix-up for fine-tuning PLMs with Bi-level Optimization (BLO). BLO refers to nested levels of optimization where the upper-level optimization performs the job of learning parameters and the lower-level optimization focuses on minimizing the training loss by modifying the weights. The authors made use of the basic idea of splitting data into training and testing data to prevent overfitting. **ASPEN** (Ye et al., 2023) proposes an approach to efficiently train for numerous tasks on a single standalone GPU with the help of a single pre-trained model being utilized for all the tasks and the method of adaptive scheduling for updation of parameters. It was noted that lightweight nature of LoRA modules promotes its usage for each task over the GPU.

Multi-LLM Agents (Shen et al., 2024) proposes Global-to-Local Progressive Fine-Tuning GLPFT strategy, comprising of two distinct stages. The first stage involves global fine-tuning, where the authors fine-tune a backbone LLM on the original training dataset without distinguishing between sub-tasks, providing the model with a comprehensive understanding of the task. After this stage, the backbone LLM is trained to sequentially output the rationale, action, and answer as introduced. The fine-tuned LLM is used to instantiate the planner, caller, and summarizer respectively, which are continually fine-tuned on respective sub-tasks.

Another domain that influences our day-to-day choices is recommendation systems. In the same domain, **TALLRec** (Bao et al., 2023) emphasizes the fact that traditionally, LLMs perform poorly in recommendation tasks due to their sole reliance on context learning. To improve accuracy of recommendation tasks, a LRLM, i.e. Large Language Recommendation Model, is fine-tuned by TALLRec. TALLRec is composed of two fine-tuning stages, alpaca tuning, the general training process for any LLMs, and rec tuning, based on the principle of instruction tuning. The model is designed to perform well under few shot training setting, resulting in whether a user would like or dislike a certain item or product depending on historical interactions. Experiments proved that TALLRec outperforms both traditional LLMs and recommendation methods, displays cross domain generalization capability.

Every domain has its challenges, especially Medicine. Whether it is the difficulty to obtain data, or the complex terminologies associated with the vast field, working with medical data is always taxing. Techniques such as **Modeling Medical Knowledge into General LLMs** from Li et al. (2023d) come to the rescue. This paper introduces AntGLM-Med-10B, which improves a 10B parameter pre-trained Language Model for medical tasks. It considers a 3-stage optimization view: one for the injection of general medical knowledge, another for fine-tuning to sub-domains, and the last for adaptation to target tasks. The datasets include QA, reasoning, multiple choice questions, and medical conversations. A novelty is Verification-of-Choice, which strengthens the handling of multi-choice questions. ESLP, while smaller in scale, outperforms larger LLMs on PubMedQA, proving to be effective in medical domains. Another study from Tan et al. (2024) investigates whether GPT-4 can be used to grade responses from fine-tuned LLM chatbots responding to ophthalmological patient questions. Ophthalmologists developed 400 questions with paired answers for training and testing, including 8 additional glaucoma questions. Five LLMs were trained and tested using a clinical rubric that included measures of accuracy, relevance, safety, and clarity. GPT-4 correlated very closely with human clinician rankings; the Spearman and Kendall Tau correlations were found to be 0.90 and 0.80, with a Cohen Kappa of 0.50. The top-scoring was GPT-3.5 at 87.1%, which proves its effectiveness in clinical alignment. This might suggest that GPT-4 could contribute a lot to the ability for automatic grading of LLM chatbot responses for health applications.

There is also been gripping interest for multimodal chatbots, following the trend is **NEXT-GPT** (Wu et al., 2023), the first end-to-end

any-to-any multimodal language-like model system. Multimodal large language models have advanced in understanding multiple modalities, text, image, video, and audio. While such models preside in one-way understanding input-side, very few of them are intended for creating content across modalities. On top of this basis, **NEXT-GPT** specifically bases an LLM, further augmented with multimodal adaptors and diffusion decoders, to process inputs and generate outputs in any modality combination. This will then expand the capacities of LLM's to support multimodality through minimal additional projection layers fine-tuned on top of the pre-trained encoders. They provide a modality-switching instruction tuning and a curated dataset to improve cross-modal semantic understanding and generation. **DALLMi** (Beġianu et al., 2024) performed extrapolation of text classification using Large Language Models across domains with altering label availability is pretty challenging. For example, movie genre classification from IMDb to Rotten Tomatoes. Most of the existing adaptation methods are less efficient in multi-label tasks. This paper proposes a new framework on semi-supervised domain adaptation for LLMs, particularly BERT, and deals with multi-label text classification. In this paper, DALLMi can effectively integrate labeled and unlabeled data by drawing on a variational loss and MixUp regularization. This model outperforms both unsupervised and partially-supervised approaches by up to 52.2% in mean average precision. This work thus constitutes a significant step toward the improvement of LLM-based classifiers over a wide range of textual domains.

Zheng et al. (2024) deals with challenges in domain-specific machine translation using large language models. It introduces LlamaIT, a fine-tuning method specifically for LLMs, where prompts are oriented to strengthen domain-specific MT efficiently. Combining mix-domain datasets specific to tasks with LoRA — language representation alignment — LlamaIT not only does not require translating input examples but also ensures that models could reduce overspecialization, thus retaining zero-shot MT capabilities. The domain-specific vocabulary is integrated into new prompts, improving translation quality for rare words. Experimental results indicate major improvements in the domain-specific MT performance with the actual realization of general-purpose MT capability preservation of LLMs.

Measuring and Reducing LLM Hallucination without Gold-Standard Answers (Wei et al., 2024) addressed the challenge of measuring, without using the requirement of gold-standard answers, hallucination in large language models and presents a new metric: Factualness Evaluations via Weighting LLMs. FEWL scores the degree of expertise that each LLM holds by how much it disagrees with model platitudes, and how deep its knowledge is about related questions. The approach offers theoretical guarantees and provides empirical improvements in accuracy over naive approaches to measuring hallucination. In addition, it has been empirically shown that methods, like in-context learning or supervised fine-tuning, were quite successful in reducing hallucination using FEWL and hence provide a cost-effective alternative to human annotation.

TAT-LLM (Zhu et al., 2024) falls under the application for solving a challenging problem of question answering over hybrid tabular and textual data, along with a special interest in documents that require discrete reasoning, such as SEC filings. This methodology uses the multi-step reasoning power of LLMs, which have been specifically designed through a stepwise pipeline: extractor to identify relevant information, reasoner to formulate logical rules, and executor to compute final answers for questions. The validation results on GPT-4 were very promising, but it seemed to be a costly and high-latency system. Therefore, the TAT-LLM model was created by fine-tuning LLaMA 2. The implication of the above is that TAT-LLM outperformed GPT-4 and other models in tabular and textual QA tasks on the FinQA, TAT-QA, and TAT-DQA benchmarks.

The ART of LLM Refinement (Shridhar et al., 2023) discusses the ability of Large Language Models to improve their generated outputs by themselves, called self-refinement. LLMs empirically seem to be

very inaccurate at detecting errors on reasoning problems. The paper now proposes ART: Ask, Refine, and Trust—a decision mechanism for deciding refinement with a significantly smaller model. ART asks specific questions to evaluate an initial LLM response, refines if necessary, and picks the better result between the refined and original predictions. It demonstrates improvements of up to 5 points over the self-refinement baselines while remaining cost-effective alternatives to fine-tuning larger models in the tasks of mathematical reasoning and question answering.

Mallows-DPO (Chen et al., 2024b) uses a variation of Direct Preference Optimization (DPO) which is one of the methods for fine-tuning LLMs using human feedback, eliminating the requirement of a reward model. DPO often assumes uniform human preference, which might turn out to be way too unrealistic to be held true in real-world scenarios. Mallows-DPO generalizes this by introducing a dispersion index that captures the variability in human preferences. This improves the flexibility and accuracy of preference optimization. The reward function in the Mallows-DPO is decomposed as a product of dispersion and scaled reward to enhance interpretability and empirical performance on a wide array of tasks.

User-LLM (Ning et al., 2024) introduces the USER-LLM framework which enhances LLMs through user embeddings that have been distilled from interaction data of great diversity. In these embeddings, user behaviors on modalities such as video watching and rating are captured by a Transformer-based encoder trained with SSL. These embeddings are then folded into the LLM through cross-attention mechanisms that adapt to users' contexts on the fly during fine-tuning. This will result in outstanding performance of the USER-LLM compared with text prompt-based methods on tasks needing deep understanding of users and handling long contexts efficiently without any loss of computational efficiency and core LLM capabilities.

4. Practical considerations

In this section, we recognize the concern of the practical difficulties of deploying fine-tuning methods in real-world application settings, especially considering the overhead in computation and the complexity of implementation. We have added sections that draw out the wide-ranging choice of methods, from lightweight approaches such as LoRA and its variants to more sophisticated techniques like DiReFT and LoReFT.

4.1. Computational overhead and efficiency trade-offs

- Techniques such as LoRA, DoRA, and MixLoRA are designed to reduce the computational costs by fine-tuning only some subset of parameters. Their effectiveness across many models, namely, LLaMA, Gemma, and GPT-J shows efficiency in real applications where retraining of full models is prohibitive.
- Full fine-tuning (FT) achieves high performance but is computationally costly since all the model parameters must be updated. This is demonstrated in the results of WebNLG, which shows that FT maintains high BLEU scores at the cost of large storage space, for instance, GPT-2 L requires 3096MB.
- Parameter-efficient tuning methods, LoRA, AlphaTuning, MixLoRA, DiReFT, and the like are very competitive even when they utilize only a fraction of trainable parameters. For example, LoRA for GPT-2 M at 0.35M params achieved 55.8 BLEU which actually surpassed FT, with greatly reduced storage and computation requirements.
- Quantization-based techniques (PTQ, QLoRA, etc.) further reduce the memory footprint but degrade the performance. Results on WebNLG show that PTQ(W+WLoRA) on GPT-2 L significantly reduced the model size but suffered in BLEU scores (e.g., 13.6).
- More complex approaches, such as DiReFT and LoReFT, although they are more accurate (for example, 95.6% for DiReFT on LLaMA-2 8B), may involve extra computation because of the architectural changes. Yet, their performance improvements are worth the trade-off for applications that value accuracy.

4.2. Implementation complexity and deployment challenges

- Some methods, such as PESC for Mistral 7B and ResLoRA variants for LLaMA-2 7B, are optimized for specific architectures and may incur additional implementation challenges. PESC requires sparsity-aware optimizations and is hardware sensitive. However, a compatibility focused design ensures better performance with minimal tuning. These adaptations improve results but require fine-tuning strategies adapted to specific model structures, which requires additional efforts in implementation and hyperparameter tuning.
- Hybrid methods (for example, LoRA + Quantization) increase the complexity in training models but enable smaller footprints during deployment. PTQ(W)+WLoRA has highly compressed models but at a cost of performance.
- Evaluation on the GSM8K dataset shows how Spectral and MoELoRA methods enhance multi-step reasoning as well, which is normally hard for traditional LLMs. These methods come with the balance of controlled complexity allowing targeted performance improvements suitable for deployment to specialized tasks. These reasoning tasks point out how autoregressive models suffer from error propagation. Spectral tuning and MoELoRA help to alleviate this, but challenges remain in real-world deployment where precision is critical.

5. Experiments

A quantitative analysis of the fine-tuning methods with respect to different fields and parameters play a pivotal role while selecting the method to be used in an application. Hence, in this section, we present an aggregation of the evaluations done by the authors of the presented works, across various benchmarks, datasets, models and metrics.

5.1. GLUE

General Language Understanding Evaluation (GLUE) (Wang et al., 2018) is a benchmark for the evaluation of language models. It consists of a collection of NLP tasks including question answering, sentiment analysis, and textual entailment, and an associated online platform for model evaluation, comparison, and analysis. This benchmark does not introduce any new datasets but uses a collection of already existing datasets and metrics which are agreed upon by the NLP community to indicate the performance of the language models in their particular domains. These datasets and metrics are :-

The Corpus of Linguistic Acceptability (CoLA) (Warstadt, 2019) is a dataset consisting of a set of sentences labeled as grammatical and ungrammatical. The task presented before the model being tested is to classify whether a sentence is grammatically correct or not. The aim of having this dataset in GLUE is to test the linguistic competence of the model. The evaluation metric reported for this task is the Matthews correlation coefficient (Matthews, 1975) that evaluates performance on unbalanced binary classification and ranges from -1 to 1 , with 0 being the performance of uninformed guessing.

The Stanford Sentiment Treebank (SST-2) (Socher et al., 2013) is the first corpus of sentiment labeled parse trees that enable a complete analysis of the compositional effect of the language on the overall sentiment of the text. For the evaluation in the GLUE benchmark, we consider sentence level sentiment. The task of the model being tested is the binary classification of the sentences to positive or negative sentiment. The metric reported for this task is the accuracy of the model during classification.

The Microsoft Research Paraphrase Corpus (MRPC) (Dolan and Brockett, 2005) is a corpus of sentence pairs created from various new data sources. These sentences were then classified to be semantically equivalent or not by a SVM-based classifier. These classifications were further verified by human judges. For this dataset, the task of the model

being tested is this classification for semantic equivalence. The metrics reported for this task is the F1 score and accuracy during classification.

Semantic Textual Similarity Benchmark (STS-B) (Yang et al., 2018) is a dataset consisting of sentence pairs which are annotated for their similarity with a score ranging from 1 to 5. The task of the model being tested is to predict this similarity score. The metrics reported for this task is the Pearson and Spearman correlation coefficients between the actual and predicted scores.

Quora Question Pairs (QQP) is a dataset consisting of pairs questions labeled for semantic equivalence. The data is obtained from the community question-answering website, Quora. The task for the model being evaluated, like MRPC, is to classify for semantic equivalence. The metrics reported for the task are the F1 scores and accuracy during classification.

The Multi-Genre Natural Language Inference Corpus (MNLI) (Williams et al., 2017) is a dataset of sentence pairs annotated for textual entailment. The task for the model being tested is to, on the basis of the premise sentence, classify the hypothesis sentence as an entailment, contradiction or neither. The metric reported for this task is the accuracy during the classification.

The Stanford Question Answering Dataset (QNLI) (Rajpurkar, 2016) is a dataset consisting of question-paragraph pairs where the answer to the question is present in one of the sentences in the paragraph. In the GLUE benchmark, the dataset is modified to pairs of question-sentences. The task of the model being tested is to classify whether the sentence contains the answer to its pairing question or not. The metric reported for the task is the accuracy during classification.

Recognizing Textual Entailment (RTE) is dataset of sentence pairs created by aggregating the datasets of multiple entailment challenges. The task of the model being tested is to classify whether one sentence entails the other. The metric reported for the task is the accuracy during the classification.

The results of the evaluation of various fine-tuning methods on the basis of this benchmark have been mentioned in Table 2. The base models used during the evaluations are DeBERTaV3 (He et al., 2021), OPT (Zhang et al., 2022b), RoBERTa (Liu, 2019), Llama-2 (Touvron et al., 2023a), Llama (Touvron et al., 2023b) and BERT (Devlin, 2018).

It can be observed that FT in general gives the best performance for most tasks, especially for DeBERTaV3. However, LoRA, and other similar efficient techniques, give equivalent results with much lesser resource consumption. The LoRA-based methods have a very strong balance between performance and efficiency due to not fully training all parameters and can be seen as a quite viable alternative to FT, though sometimes these may lead to slight reductions in accuracy for tasks like CoLA or STS-B. It is also observed that OPT models underperform on classification tasks, mostly because they are optimized for generation and not bidirectional understanding.

5.2. E2E

Based on the E2E challenge, this method is used to measure the end-to-end, data-driven natural language generation capabilities of a model. The challenge uses the E2E dataset (Novikova et al., 2017) which was a crowd sourced dataset that was 10 times the size of its predecessors in the domain. The dataset showed more lexical richness and syntactic variation, including discourse phenomena. The difficulty in performing on this dataset also lies in the fact that it enforces content selection where the model is also tested for the task of selecting the content to be realized. This is because while generating this dataset, the our crowd workers were asked to verbalize all the useful information from the and were allowed to skip an attribute value considered unimportant.

The metrics used to measure the performance of the model on this dataset as per the E2E challenge are BLEU (Papineni et al., 2002), NIST (Doddington, 2002), METOEOR (Banerjee and Lavie, 2005), ROUGE_L (Lin, 2004) and CIDEr (Vedantam et al., 2015). BLEU measures the performance of text generation in terms of the precision

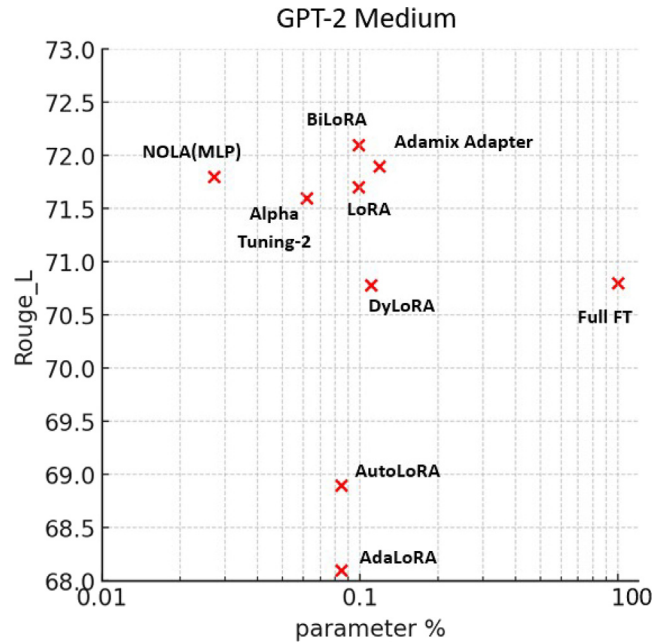


Fig. 2. Scatter plot of Rouge_L scores of various models against parameter percentage.

with respect to the n-grams in the generated text. NIST uses a similar methodology but it weighs more informative n-grams heavily. METEOR takes into consideration the presence of synonymous words and uses the stem form of words during evaluation. ROUGE_L evaluates longest common subsequence recall and precision. CIDEr focuses on consensus using term frequency-inverse document frequency (TF-IDF) weighting.

The results of the evaluation of various fine-tuning methods on the basis of this challenge have been mentioned in Table 3. The models from the GPT-2[129] family have been used as base models for evaluation.

Notably, LoRA and AlphaTuning show better performance on BLEU, NIST, METEOR, ROUGE_L, and CIDEr while requiring significantly fewer trainable parameters for being efficient substitutes for full fine-tuning. They are more effective in adaptively updating the key parameters while maintaining the pre-trained knowledge and optimizing it according to particular tasks. However, when PTQ methods are applied with LoRA, there is a noticeable degradation in performance, especially in CIDEr, which heavily relies on semantic richness and originality—both of which can be degraded by aggressive quantization. AlphaTuning performs equally well as LoRA for GPT-2 M and GPT-2 L, and it is able to adapt to different model sizes without any excessive computational cost.

Additionally, Fig. 2 represents a scatter plot where the Rouge_L score is plotted against percentage of parameters being trained, over GPT-2 Medium model.

5.3. MMLU

Massive Multitask Language Understanding (MMLU) (Hendrycks et al., 2020) is a benchmark used to measure the multitask accuracy of the model. The benchmark contains tasks of up to 57 different domains at varying difficulties including Humanities, Social Science, STEM and others. The expansive nature of this benchmark, in terms of the domains of the tasks, helps researchers find aspects in which their methods maybe lacking.

The results of the evaluation of various fine-tuning methods on the basis of this benchmark have been mentioned in Table 4. The models from the Llama and Llama-2 families have been used as base models for evaluation.

Table 2

Evaluation of Fine-Tuning methods as per the GLUE Benchmark.

Base model	Method	Tasks							
		Single sentence		Similarity and paraphrase			Inference		
		CoLA	SST-2	MRPC	STS-B	QQP	MNLI	QNLI	RTE
		Matt	Acc	F1/Acc	P/S Corr	F1/Acc	Acc	Acc	Acc
DeBERTaV3	Full FT	69.2	95.3	89.5/93.3	91.6/91.1	92.4/89.8	90.6	94.0	82.0
	LoRA	69.9	95.1	89.9/93.6	91.7/91.1	92.0/89.4	90.5	94.6	85.1
	QLoRA	–	86.5	73.8/82.8	83.0/82.8	86.8/82.3	75.6	82.4	55.9
	LoftQ	37.4	90.2	83.8/88.6	87.1/86.9	90.3/86.9	85.1	86.6	61.4
	IncreLoRA	71.8	96.2	-/91.0	91.9/-	-/92.2	90.9	94.4	88.2
	BOFT	72.9	96.4	-/92.4	91.9/-	-/92.1	90.2	94.2	88.8
	PRILoRA	72.8	96.2	-/92.5	91.9/-	-/92.4	90.7	94.4	89.1
	AdaLoRA	71.4	96.1	-/90.69	91.84/-	89.74/92.23	90.7	94.5	88.1
	AFLoRA	72.0	96.2	-/91.9	91.8/-	90.8/87.8	89.8	94.4	88.1
	SpectralAdapter	71.9	96.7	-/91.7	91.5/-	-/91.9	90.4	94.3	88.1
	WeightMixup	65.9	–	92.32/-	-/91.9	–	–	–	86.1
OPT-1.3B	LoRA	62.5	93.7	- /83.4	89.3/ -	- /86.9	81.2	81.4	82.7
	QLoRA	62.1	94.9	- /83.3	89.9/ -	- /87.1	76.0	86.3	81.3
	QST	59.7	94.3	- /81.7	88.4/ -	- /84.3	77.1	85.7	79.5
OPT-2.7B	LoRA	64.5	95.3	- /84.6	90.9/ -	- /90.7	82.6	83.0	84.7
	QLoRA	63.7	95.6	- /84.8	91.2/ -	- /89.5	78.3	88.7	83.6
	QST	62.0	95.2	- /83.7	88.9/ -	- /86.5	80.4	86.6	80.1
OPT-6.7B	QLoRA	64.3	96.2	- /85.9	92.0/ -	- /90.7	79.8	90.2	84.5
	QST	62.8	96.4	- /85.2	89.6/ -	- /89.4	81.6	87.3	80.8
RoBERTa	LoRA	66.1	94.5	92.3/ -	90.9/ -	90.5/ -	87.0	92.8	78.0
	DyLoRA	63.3	94.7	92.8/ -	91.4/ -	90.2/ -	87.2	93.6	80.1
	SMART	70.6	96.9	89.2/92.1	92.8/92.6	92.4/89.8	91.1	95.6	92.0
	LoRA SIBO	68.5	96.4	90.4/-	-/92.4	-/91.6	90.5	95.0	85.2
	Adapter SIBO	67.1	96.6	90.8/-	-/92.1	-/91.9	89.9	94.8	84.8
	MEFT ₁ (Vanilla)	69.9	96.8	-/91.5	92.3/-	-/91.5	90.3	94.9	89.5
	LoReFT	68.0	96.2	-/90.1	91.6/-	88.5	-/89.2	94.1	87.5
	ResLoRA _{ss}	65.4	96.2	91.3/-	91.7/-	91.3/-	90.4	94.5	82.3
	ColA(MLP)	64.6	95.1	-/90.4	90.7/-	-/91.3	86.9	92.4	73.6
	AdaMix	70.2	97.1	-/91.9	92.4/-	-/92.3	90.9	95.4	89.2
	AutoLoRA	61.3	94.9	-/89.4	-/90.8	-/90.3	87.0	92.9	77.0
	Hydra	65.4	95.0	-/92.2	91.7/-	-/90.8	87.5	92.8	87.4
	HyPe-U	65.8	–	92.6/-	92.3/-	–	–	–	84.1
	Child-Tuning _D	66.7	–	92.8/-	-/92.3	–	–	–	86.1
	WeightMixup	66.5	–	92.6/-	/92.2	–	–	–	84.2
LlaMA-2 7B	LoRA	–	94.1	–	–	–	–	93.1	83.3
	AdaLoRA	–	94.1	–	–	–	–	93.2	83.5
	ALoRA	–	95.0	–	–	–	–	93.7	84.6
BERT-Base	SMART	59.1	93.0	87.7/91.3	90.0/89.4	91.5/88.5	85.6	91.7	71.2
	GamMa	60.0	92.3	88.9/-	–	87.6/-	83.3	90.5	71.8
	MEFT ₁ (Vanilla)	62.1	92.9	-/86.7	89.0/-	-/89.9	83.8	91.6	74.2
	Adamix	62.9	92.4	92.4/89.5	89.9/-	87.6/90.7	84.7	91.5	74.7
BERT-Large	DPS Mix	65.1	–	91.8/-	-/90.5	–	–	–	73.5
	LoRA SIBO	61.6	93.8	90.8/-	-/89.9	-/87.7	85.6	92.2	71.8
	Adapter SIBO	63.1	94.6	90.9/-	-/90.2	-/88.3	86.0	92.4	73.2
	GamMa	65.1	94.5	91.5/-	–	87.3/-	86.8	93.9	74.3
	Child Tuning _D	64.9	–	91.4/-	-/90.2	–	–	–	73.1
	HyPe-U	65.5	–	92.1/-	-/90.3	–	–	–	74.5
	WeightMixup	66.1	–	91.8/-	-/90.3	–	–	73.4	–

Table 3

Evaluation of Fine-Tuning methods on the E2E Dataset.

Evaluation metrics									
Model	Method	q	Trainable params	Valid Loss	BLEU	NIST	METEOR	ROUGE_L	CIDEr
GPT-2 M	Fine-Tuning	–	354.92M	1.28	67.5±.2	8.60±.02	46.4±.2	70.8±.2	2.40±.01
	PTQ(WFT)	3	–	1.19	67.5±.5	8.58±.04	46.3±.5	70.3±.1	2.39±.01
	LoRA	–	0.35M	1.16	70.2±.2	8.80±.04	46.8±.1	71.7±.4	2.53±.01
	PTQ(W)+WLoRA	3	–	4.10	11.1 ± 5.3	2.35 ± 1.04	12.4 ± 4.2	29.9 ± 7.8	0.35±.18
	PTQ(W)+WLoRA)	3	–	4.38	7.5 ± 3.2	1.31±.21	11.4±.8	29.8 ± 3.0	0.29±.04
	AlphaTuning	3	0.22M	1.18	69.9±.3	8.79±.05	46.7±.2	71.7±.3	2.51±.01
	AlphaTuning	2	0.22M	1.20	70.0±.4	8.80±.05	46.7±.1	71.6±.5	2.51±.01
GPT-2 L	Fine-Tuning	–	774.03M	1.31	67.2±.3	8.61±.05	46.3±.1	70.5±.3	2.37±.01
	PTQ(WFT)	3	–	0.98	66.5 ± 1.0	8.45±.10	45.7±.3	70.3±.5	2.37±.03
	LoRA	–	0.77M	1.13	69.8±.2	8.80±.03	46.6±.1	71.7±.1	2.51±.01
	PTQ(W)+WLoRA	3	–	1.87	50.9 ± 3.4	6.63±.32	38.7 ± 1.9	60.6 ± 1.9	1.30±.19
	PTQ(W)+WLoRA)	3	–	1.76	53.7 ± 3.1	7.12±.37	40.0 ± 1.5	61.7 ± 1.1	1.5±.11
	AlphaTuning	2	0.42M	1.14	69.7±.6	8.78±.08	46.6±.2	71.5±.3	2.51±.03
	AlphaTuning	1	0.42M	1.18	69.7±.3	8.79±.03	46.6±.1	71.6±.2	2.51±.02

Table 4
Evaluation of Fine-Tuning methods on the major tasks of the MMLU Benchmark.

Method	Tasks				
	Hums.	Social	STEM	Other	Avg.
LLaMA-7B fine-tuned on Alpaca					
Without FT	33.3	37.8	29.8	38.0	34.6
QLoRA	36.1	42.0	31.9	44.5	38.4
QA-LoRA	36.6	44.8	32.4	44.9	39.4
IR-QLoRA	38.6	45.2	34.6	45.5	40.8
LLaMA-7B fine-tuned on Flan v2					
QLoRA	41.4	49.8	35.0	52.0	44.3
QA-LoRA	43.9	54.3	38.0	53.0	47.0
IR-QLoRA	44.2	54.5	39.3	52.9	47.4
LLaMA-7B fine-tuned on Tulu					
FFT	38.8	45.8	34.4	47.8	41.3
LoRA	37.6	47.7	33.9	47.8	41.2
Periodic LoRA	39.0	45.8	33.8	48.2	41.4
LLaMA2-7B fine-tuned on Alpaca					
Without FT	43.0	51.4	36.4	52.2	45.5
QA-LoRA	42.1	49.1	34.4	50.3	43.9
IR-QLoRA	43.4	51.9	36.8	53.6	46.2
LLaMA2-7B fine-tuned on Flan v2					
QA-LoRA	48.4	59.4	41.4	58.6	51.7
IR-QLoRA	49.2	60.2	41.6	58.0	52.0
LLaMA-13B fine-tuned on Alpaca					
Without FT	40.6	48.9	36.7	48.0	43.3
QLoRA	45.4	55.7	37.4	54.3	48.0
QA-LoRA	48.4	54.9	38.3	55.2	49.2
IR-QLoRA	47.2	56.5	39.0	55.0	49.3
LLaMA-13B fine-tuned on Flan v2					
QLoRA	49.9	60.2	40.1	57.9	51.9
QA-LoRA	50.0	60.5	41.5	58.4	52.4
IR-QLoRA	49.2	62.1	41.2	59.2	52.6
LLaMA2-13B fine-tuned on Alpaca					
Without FT	53.3	63.3	44.1	61.0	55.3
QA-LoRA	48.0	59.7	43.0	57.4	51.7
IR-QLoRA	51.9	61.9	43.9	60.4	54.4
LLaMA2-13B fine-tuned on Flan v2					
QA-LoRA	52.9	65.9	44.8	64.0	56.6
IR-QLoRA	53.1	64.9	45.6	63.8	56.5

The table presents the following trends regarding fine-tuning methods in relation to various sizes of LLaMA and LLaMA2. First, the full fine-tuning technique normally achieves the highest performance; however, parameter-efficient techniques like QLoRA, QA-LoRA, and IR-QLoRA achieve competitive performance at reduced computational costs. Overall, IR-QLoRA performed better than other LoRA-based techniques across all tasks, which may indicate that adding retrieval-augmented adaptation aids in better recall of knowledge and reasoning. Additionally, models fine-tuned on Flan v2 outperform those fine-tuned on Alpaca, likely due to Flan v2's richer and more diverse instruction-tuning dataset, which enhances generalization across subjects. However, LoRA-based methods are just slightly underperforming in the tasks of STEM, indicating that full fine-tuning could be more beneficial for tasks with deep numerical and logical reasoning.

5.4. HellaSwag

HellaSwag (Zellers et al., 2019) is an extension of the SWAG dataset (Situations with Adversarial Generations) designed for commonsense natural language inference. This dataset contains sentences along with options for its possible continuation. The options are designed in an adversarial manner, so to understand the difference between them and picking the correct continuation requires deep language understanding and generalization abilities. This dataset proved that commonsense

Table 5
Comparison of fine-tuning methods on the basis of HellaSwag dataset for zero-shot.

Base model	Method	Accuracy
Llama 7B	LoRA Adapter	78.1
	Series Adapter	67.9
	Parallel Adapter	69.8
	IR-QLoRA	54.7
	QA LoRA	58.6
	OLoRA	77.86
	DiReFT	92.5
	LoReFT	93.1
	PEQA	71.4
	DLoRA	73.6
Llama 13B	LoRA Adapter	90.5
	Series Adapter	88.1
	Parallel Adapter	92.1
	DiReFT	94.6
	LoReFT	95.1
Mistral 7B	PeQA	74.0
	Mistral 7B	81.3
Llama-2 7B	PESC	86.5
	DiReFT	93.6
	LoReFT	94.3
	ResLoRA _{fs}	62.3
	ResLoRA _{bs}	72.1
	ResLoRA _{ms}	88.2
	LoRA	91.9
	DoRA	89.6
	MixLoRA	93.1
	MixDoRA	93.1
Llama-2 8B	TiedLoRA ₆	91.9
	DiReFT	95.6
Llama-2 13B	LoReFT	96.3
	LoRA	94.3
	DoRA	94.8
	MixLoRA	94.7
GPT-J 6B	MixDoRA	95.2
	Adapter	30.6
	LoRA	43.1
	Adapter SIBO	48.7
Gemma 2B	LoRA SIBO	47.8
	LoRA	84.4
	DoRA	85.4
	MixLoRA	89.0
	MixDoRA	89.3

inference is still difficult even for state-of-the-art models. This is due to the adversarial nature of the data and only minor differences between plausible and implausible sentences.

The results of the evaluation of various fine-tuning methods on the basis of this dataset have been mentioned in Table 5. The base models used during the evaluations are Llama, Mistral, Llama-2, GPT-J (Wang and Komatsuzaki, 2021) and Gemma (Team et al., 2024).

Performance trends reveal that the most advanced methods such as DiReFT, LoReFT, and Mix-based methods are successful because they are more adaptable to model architecture and scale, especially for larger models like LLaMA-2. Simpler techniques, like QA LoRA and IR-QLoRA, underperform on smaller models, reflecting their limited optimization capacity. Architecture-specific methods, like PESC for Mistral 7B, demonstrate the importance of compatibility with the model's structure. Poor performance for GPT-J 6B indicates architectural limitations, but optimized smaller models like Gemma 2B show that the right fine-tuning strategy can still achieve competitive performance. Fine-tuning success depends on the sophistication of the method and model compatibility.

5.5. GSM8K

Grade School Math 8k (GSM8k) (Cobbe et al., 2021) is a dataset consisting of high quality linguistically diverse grade school math problems

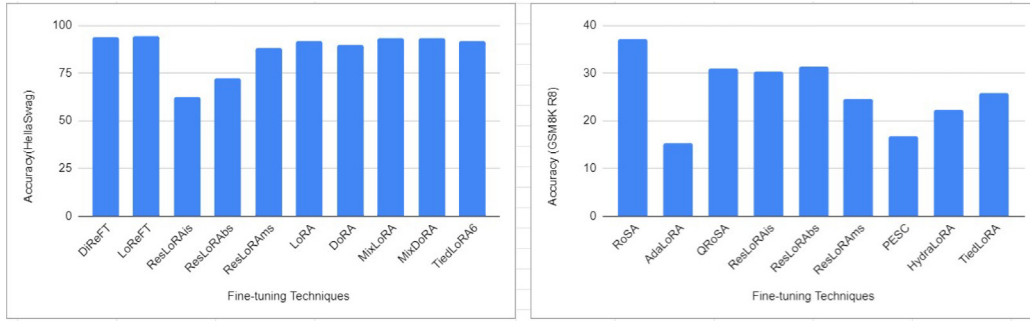


Fig. 3. Bar graphs representing scores of models on HellaSwag dataset (left) and GSM8K (right) over Llama-2 7B model.

Table 6

Comparative analysis of fine-tuning methods on GSM8K dataset (zero-shot) with accuracies measured across various ranks.

Base model	Method	Accuracy		
		R8	R16	R32
LlaMA 7B	LoRA	20.6	–	37.5
	DyLoRA	19.94	18.50	–
	QDyLoRA	16.76	16.91	–
	PLoRA	24.0	–	–
	DiReFT	24.1	–	–
	LoReFT	26.0	–	–
	AFLoRA	–	–	38.6
LlaMA 13B	LoRA	47.5	–	–
	DiReFT	35.8	–	–
	LoReFT	38.1	–	–
LlaMA-2 7B	LoRA	–	32.9	50.1
	RoSA	37.2	38.4	36.2
	AdaLoRA	15.3	–	–
	QLoRA	–	29.8	–
	QRoSA	30.9	33.1	32.3
	ResLoRA _{is}	30.3	–	–
	ResLoRA _{bs}	31.3	–	–
	ResLoRA _{ms}	24.6	–	–
	OFT	–	–	50.1
	BOFT	–	–	50.6
	PESC	16.7	–	–
	HydraLoRA	22.3	–	–
	TiedLoRA	25.7	27.1	–
Falcon 40B	QLoRA	15.1	20.5	–
	QDyLoRA	30.6	29.8	–
GPT 3.5	Series Adapter	–	–	18.5
	Parallel Adapter	–	–	22.7
	LoRA	–	–	31.1
	MoELoRA	–	–	32.6
Mistral 7B	LoRA	46.2	–	–
	DoRA	45.8	–	–
	Spectral	51.1	–	–

created by human problem writers. This dataset was created to enable research on multi-step mathematical reasoning tasks on which even the state-of-the-art models have consistently given poor performance. This is because autoregressive models have no mechanisms to correct themselves, hence, even a minor mistake cascades down to further steps, making the entire solution incorrect.

The results of the evaluation of various fine-tuning methods on the basis of this dataset have been mentioned in Table 6. The ranks of the adapters taken into consideration while evaluating the models are 8, 16 and 32. The base models used during the evaluations are Llama, Mistral, Llama-2, GPT-3.5 (Brown, 2020) and Falcon-40B (Almazrouei et al., 2023). Fig. 3 represents a comparison of various approaches in form of bar graph.

Additionally, Table 6 indicates that LoRA-based methods are still effective, but enhanced variants such as LoReFT, DiReFT, and Spectral

fine-tuning often outperform standard LoRA, indicating that adaptation technique refinement improves accuracy. Quantized methods (e.g., QLoRA, QDyLoRA) tend to perform worse, likely because precision loss impacts numerical reasoning. Higher-rank settings tend to yield better results, as seen with LLaMA-2 7B and GPT-3.5, reinforcing the importance of parameter capacity. Additionally, Spectral fine-tuning is seen to greatly improve the accuracy of Mistral 7B, highlighting the role of architectural compatibility in fine-tuning.

5.6. WEBNLG

The WebNLG challenge (Gardent et al., 2017) provided a benchmark to evaluate and compare microplanners. Microplanners are components involved in the text generation process which include tasks like content selection which involves selecting the information which has to be realized, lexicalization which involves the choice of specific words or phrases to represent the information and aggregation which involves combining multiple pieces of information to create a coherent sentence. The dataset for this challenge consisted of a mapping set of Resource Description Framework (RDF) triples to text. A RDF triple consists of a subject which is an entity, a predicate which is a property of that entity and an object which is the value of that property.

The results of the evaluation of various fine-tuning methods on the basis of this dataset have been mentioned in Table 7. The models from the GPT-2 family have been used as base models for evaluation. The comparison highlights that FT has the highest performance but incurs the maximum trainable parameters; thus, it is a very resource-intensive technique. LoRA achieves the comparable BLEU scores with a minimum trainable parameter, showing efficiency and adaptability. The different levels of quantization (q) in AlphaTuning maintain an effective trade-off between performance and resource utilization with competitive BLEU scores and reduced model size, particularly at higher levels of quantization. The PTQ-based approaches, like PTQ(WFT), PTQ(W)+WLoRA, have remarkable resource efficiency through size-reduced models but incur high validation loss along with BLEU scores that degraded to some extent, thus giving a trade-off between compression and quality.

5.7. Medical benchmarks

Application specific research has been done in medical fields by fine-tuning LLMs in a specific manner to cater to the medical industry needs. One such dataset benchmark in this field is PubMedQA (Jin et al., 2019). It is a biomedical QA dataset collected from PubMed abstracts. The answers for the questions has been pre-configured as yes/no/maybe. The results for the various domain specific models trained for medical applications with their different ranks has been documented in Table 8.

Table 7

Evaluation of Fine-Tuning methods on the WebNLG Dataset.

Model	Method	Evaluation metrics							
		q	Trainable params	Model size		Valid loss	BLEU (95% Confidence interval)		
			Params	CKPT	Total		Unseen	Seen	All
GPT-2 M	FT (Fine-Tuning)	–	354.9M	1420MB	1420MB	0.79	32.7±.6	62.0±.4	48.4±.3
	PTQ(WFT)	3	–	327MB	327MB	2.03	25.0 ± 2.5	58.7 ± 1.0	43.2 ± 3.3
	LoRA	–	0.35M	1.4MB	1420MB	0.81	45.5±.4	64.3±.2	55.8±.3
	PTQ(W)+WLoRA	3	–	1.4MB	328MB	2.98	15.8 ± 3.0	15.8 ± 3.4	15.8 ± 3.2
	PTQ(W+WLoRA)	3	–	327MB	327MB	3.36	12.6 ± 4.1	16.6 ± 6.7	13.6 ± 7.5
	AlphaTuning	3	0.22M	0.9MB	327MB	0.81	40.9±.5	63.2±.5	53.1±.4
	AlphaTuning	2	0.22M	0.9MB	289MB	0.84	37.3±.5	62.6±.5	51.3±.5
GPT-2 L	FT (Fine-Tuning)	–	774.0M	3096MB	3096MB	0.81	23.8±.3	60.8±.1	43.0±.3
	PTQ(WFT)	3	–	535MB	535MB	1.90	23.2±.8	62.7±.2	43.7±.7
	LoRA	–	0.77M	3.1MB	3096MB	0.79	48.4±.3	64.0±.3	57.0±.1
	PTQ(W)+WLoRA	3	–	3.1MB	538MB	1.97	20.1 ± 5.2	27.8 ± 4.1	24.1 ± 4.5
	PTQ(W+WLoRA)	3	–	535MB	535MB	1.97	14.0 ± 7.2	26.6 ± 11.5	25.8 ± 13.0
	AlphaTuning	3	0.42M	1.7MB	535MB	0.84	47.0±.6	62.2±.2	55.3±.3
	AlphaTuning	2	0.42M	1.7MB	445MB	0.82	42.7±.4	62.9±.4	53.8±.1
	AlphaTuning	1	0.42M	1.7MB	355MB	0.87	28.1±.3	62.3±.7	47.1±.4

Table 8

Evaluation of Fine-Tuning methods on the PubMed Dataset.

Model	r	PubMedQA
FFT	–	74.1
LoRA	8	73.1
MoRA	8	73.3
AntGLM-Med-10B	8	80.6
LoRA	256	73.8
MoRA	256	74.4

Table 9

Comparative analysis of fine-tuning methods on OpenBookQA.

Base model	Techniques	Accuracies
LLaMA-3-8B	LoRA	85.00
	DoRA	80.60
	MixLoRA	85.80
	MixDoRA	86.90
OPT-1.3B	LoRA (32)	22.00
	OLoRA (32)	23.00
	LoRA (64)	21.20
	OLoRA (64)	23.80
	MEFT1	34.00
	Full FT	31.40
OPT-6.7B	Zero-Shot	27.60
	MEFT1	37.00
Tiny-Llama-1.1B	LoRA (32)	35.60
	OLoRA (32)	36.00
	LoRA (64)	35.40
	OLoRA (64)	36.00
Gemma-2B	LoRA	71.40
	DoRA	70.40
	MixLoRA	75.80
	MixDoRA	75.40
LLaMA-2-7B	LoRA (32)	45.00
	OLoRA (32)	46.40
	LoRA (64)	45.00
	OLoRA (64)	46.80
	FT	70.00
	DLoRA	73.80
	DoRA	80.60
	MixLoRA	81.60
	MixDoRA	80.90
LLaMA-2-13B	LoRA	83.20
	DoRA	84.50
	MixLoRA	83.00
	MixDoRA	83.40
Mistral-7B	LoRA (32)	33.80
	OLoRA (32)	34.20
	LoRA (64)	44.60
	OLoRA (64)	45.20

5.8. Open book QA

OpenBookQA is a new kind of question-answering dataset modeled after open book exams for assessing human understanding of a subject. It consists of 5957 multiple-choice elementary-level science questions which probe the understanding of a small book of 1326 core science facts and the application of these facts to novel situations. Mihaylov et al. (2018) The results for the various models trained using different fine-tuning techniques have been documented in Table 9.

6. Conclusion and future work

This review provides an in-depth overview of the fine-tuning methods applied to a wide variety of tasks. These methods are important because they enable large language models to be more adaptable and efficient, and help them in addressing specific downstream tasks with more precision. These fine-tuning approaches have been summarized into six primary categories: Adapter Changes, Parameter Selection, Mixture of Experts (MoE), Quantization, Training Method Modifications, and Application-Specific Techniques. Each of these methods reflects the continued progress in the fine-tuning of large language models, of which MoE recently drew special attention due to its scalability, which is extremely important as models grow larger and require more efficient use of computational resources.

A detailed comparative analysis of the methods across multiple benchmarks and datasets demonstrates the rapid improvement from the basic full fine-tuning methods, not only terms of memory but also in terms of performance. This indicates that although the fine-tuning is the performance gold standard, parameter-efficient fine-tuning methods such as LoRA and its advanced versions like AsyLoRA, MoRA, IR-QLoRA are competitive for efficiency at the cost of some accuracy loss. In the knowledge retention direction, retrieval-augmented approaches, and for models like Mistral 7B, spectral fine-tuning enhances their performance, further stressing the need for architectural compatibility. Our analysis shows that there is still some loss of semantic richness

even when quantization-aware fine-tuning such as PTQ+LoRA is applied and thus an advanced loss-aware strategy for quantization is in high demand. Going forward, more performance-optimal hybrid fine-tuning strategies with the spectral-retrieval techniques could be followed to achieve further gains on STEM and continual learning tasks. In addition, model-specific adaptation frameworks for different architectures, such as PESC for Mistral 7B may help reduce adaptation efficiency losses. These insights help to explore more flexible

and resource-efficient fine-tuning strategies in the next generation of language models.

Looking forward, innovations in fine-tuning techniques will make model improvements with greater accuracy and wider applicability across domains. A continuous improvement in memory and optimization using these methods also ensures the achievement of even greater levels of model customization and adaptability, which are key factors needed to meet the changing demands in the industry. Ultimately, this review offers another perspective on recent fine-tuning trends and encourages further research in model fine-tuning.

CRedit authorship contribution statement

Samar Pratap: Writing – original draft. **Alston Richard Aranha:** Writing – original draft. **Divyanshu Kumar:** Writing – original draft. **Gautam Malhotra:** Writing – original draft. **Anantharaman Palacode Narayana Iyer:** Supervision. **Shylaja S.S.:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Agiza, A., Neseem, M., Reda, S., 2024. MTLORA: Low-rank adaptation approach for efficient multi-task learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 16196–16205.
- Almazrouei, E., Alobeidli, H., Alshamsi, A., Cappelli, A., Coccaro, R., Debbah, M., Goffinet, E., Heslow, D., Launay, J., Malartic, Q., Noun, B., Pannier, B., Penedo, G., 2023. Falcon-40B: An open large language model with state-of-the-art performance.
- Amari, S.I., 1993. Backpropagation and stochastic gradient descent method. *Neurocomputing* 5 (4–5), 185–196.
- Bai, J., Chen, D., Qian, B., Yao, L., Li, Y., 2024. Federated fine-tuning of large language models under heterogeneous language tasks and client resources. arXiv preprint arXiv:2402.11505.
- Baldazzi, T., Bellomarin, L., Ceri, S., Colombo, A., Gentili, A., Sallinger, E., 2023. Fine-tuning large enterprise language models via ontological reasoning. In: International Joint Conference on Rules and Reasoning. Springer Nature Switzerland, Cham, pp. 86–94.
- Banerjee, S., Lavie, A., 2005. METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization. pp. 65–72.
- Bao, K., Zhang, J., Zhang, Y., Wang, W., Feng, F., He, X., 2023. Tallrec: An effective and efficient tuning framework to align large language model with recommendation. In: Proceedings of the 17th ACM Conference on Recommender Systems. pp. 1007–1014.
- Bejtian, M., Mälan, A., Aldinucci, M., Birke, R., Chen, L., 2024. DALLMi: Domain adaptation for LLM-based multi-label classifier. In: Pacific-Asia Conference on Knowledge Discovery and Data Mining. Springer Nature Singapore, Singapore, pp. 277–289.
- Beltagy, I., Peters, M.E., Cohan, A., 2020. Longformer: The long-document transformer. arXiv preprint arXiv:2004.05150.
- Benedek, N., Wolf, L., 2024. PRILORA: pruned and rank-increasing low-rank adaptation. arXiv preprint arXiv:2401.11316.
- Blum, S.M., 2023. Investigating Biases in Rules Extracted from Language Models (Master's thesis). The University of Bergen.
- Brown, T.B., 2020. Language models are few-shot learners. arXiv preprint arXiv:2005.14165.
- Büyükkay, K., 2024. OLoRA: orthonormal low-rank adaptation of large language models. arXiv preprint arXiv:2406.01775.
- Cai, W., Jiang, J., Qin, L., Cui, J., Kim, S., Huang, J., 2024a. Shortcut-connected expert parallelism for accelerating mixture-of-experts. arXiv preprint arXiv:2404.05019.
- Cai, T., Li, Y., Geng, Z., Peng, H., Lee, J.D., Chen, D., Dao, T., 2024b. Medusa: Simple llm inference acceleration framework with multiple decoding heads. arXiv preprint arXiv:2401.10774.
- Chavan, A., Liu, Z., Gupta, D., Xing, E., Shen, Z., 2023. One-for-all: Generalized lora for parameter-efficient fine-tuning. arXiv preprint arXiv:2306.07967.
- Chen, Z., Deng, Y., Yuan, H., Ji, K., Gu, Q., 2024a. Self-play fine-tuning converts weak language models to strong language models. arXiv preprint arXiv:2401.01335.
- Chen, Y., Qian, S., Tang, H., Lai, X., Liu, Z., Han, S., Jia, J., 2023. Longlora: Efficient fine-tuning of long-context large language models. arXiv preprint arXiv:2309.12307.
- Chen, H., Zhao, H., Lam, H., Yao, D., Tang, W., 2024b. Mollows-DPO: fine-tune your LLM with preference dispersions. arXiv preprint arXiv:2405.14953.
- Cho, K., 2014. On the properties of neural machine translation: Encoder-decoder approaches. arXiv preprint arXiv:1409.1259.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., et al., 2021. Training verifiers to solve math word problems. arXiv preprint arXiv:2110.14168.
- Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L., 2024. Qlora: Efficient finetuning of quantized llms. *Adv. Neural Inf. Process. Syst.* 36.
- Devlin, J., 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805.
- Diao, E., Le, Q., Wu, S., Wang, X., Anwar, A., Ding, J., Tarokh, V., 2024. CoLA: Collaborative adaptation with gradient learning. arXiv preprint arXiv:2404.13844.
- Ding, Y., Zhang, L.L., Zhang, C., Xu, Y., Shang, N., Xu, J., et al., 2024. Longrope: Extending llm context window beyond 2 million tokens. arXiv preprint arXiv:2402.13753.
- Doddington, G., 2002. Automatic evaluation of machine translation quality using n-gram co-occurrence statistics. In: Proceedings of the Second International Conference on Human Language Technology Research. pp. 138–145.
- Dolan, B., Brockett, C., 2005. Automatically constructing a corpus of sentential paraphrases. In: Third International Workshop on Paraphrasing. IWP2005.
- Dou, S., Zhou, E., Liu, Y., Gao, S., Shen, W., Xiong, L., et al., 2024. LoRAMoE: Alleviating world knowledge forgetting in large language models via MoE-style plugin. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 1932–1945.
- Elman, J.L., 1990. Finding structure in time. *Cogn. Sci.* 14 (2), 179–211.
- Gao, C., Zhang, S.Q., 2024. Dora: Distributed parameter-efficient fine-tuning solution for large language model. arXiv preprint arXiv:2404.05182.
- Gardent, C., Shimorina, A., Narayan, S., Perez-Beltrachini, L., 2017. The WebNLG challenge: Generating text from RDF data. In: 10th International Conference on Natural Language Generation. pp. 124–133. ACL Anthology.
- Ge, Z., Huang, H., Zhou, M., Li, J., Wang, G., Tang, S., Zhuang, Y., 2024. Worldgpt: Empowering llm as multimodal world model. arXiv preprint arXiv:2404.18202.
- Graves, A., Graves, A., 2012. Long short-term memory. In: Supervised Sequence Labelling with Recurrent Neural Networks. pp. 37–45.
- Gui, A., Ye, J., Xiao, H., 2023. GamMa: Efficient fine-tuning of pre-trained language models using gradient activation mapping masking. In: 2023 International Joint Conference on Neural Networks. IJCNN, IEEE, pp. 1–8.
- Gui, A., Ye, J., Xiao, H., 2024. G-adapter: Towards structure-aware parameter-efficient transfer learning for graph transformer networks. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, no. 11, pp. 12226–12234.
- Guo, H., Greengard, P., Xing, E.P., Kim, Y., 2023. Lq-lora: Low-rank plus quantized matrix decomposition for efficient language model finetuning. arXiv preprint arXiv:2311.12023.
- Guo, S., Wu, F., Zhang, L., Zheng, X., Zhang, S., Chao, F., et al., 2024. EBFT: Effective and block-wise fine-tuning for sparse LLMs. arXiv preprint arXiv:2402.12419.
- Havrilla, A., Raparthy, S., Nalmpantis, C., Dwivedi-Yu, J., Zhuravinskiy, M., Hambro, E., Railneau, R., 2024. Glore: When, where, and how to improve llm reasoning via global and local refinements. arXiv preprint arXiv:2402.10963.
- Hayou, S., Ghosh, N., Yu, B., 2024. Lora+: Efficient low rank adaptation of large models. arXiv preprint arXiv:2402.12354.
- He, P., Gao, J., Chen, W., 2021. DeBERTaV3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. arXiv preprint arXiv:2111.09543.
- He, Z., Zhu, J., Qian, S., Chai, J., Koutra, D., 2024. LinkGPT: teaching large language models to predict missing links. arXiv preprint arXiv:2406.04640.
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., Steinhardt, J., 2020. Measuring massive multitask language understanding. arXiv preprint arXiv:2009.03300.
- Houlsby, N., Giurugu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., et al., 2019. Parameter-efficient transfer learning for NLP. In: International Conference on Machine Learning. PMLR, pp. 2790–2799.
- Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., et al., 2021. Lora: Low-rank adaptation of large language models. arXiv preprint arXiv:2106.09685.
- Hu, Z., Wang, L., Lan, Y., Xu, W., Lim, E.P., Bing, L., et al., 2023. Llm-adapters: An adapter family for parameter-efficient fine-tuning of large language models. arXiv preprint arXiv:2304.01933.
- Hui, T., Zhang, Z., Wang, S., Xu, W., Sun, Y., Wu, H., 2024. HFT: half fine-tuning for large language models. arXiv preprint arXiv:2404.18466.
- Hwang, I., Park, H., Yang, J., Maeng, S., Lee, Y., 2024. PC-LoRA: Progressive model compression with low rank adaptation. In: 5th Workshop on Practical ML for Limited/Low Resource Settings.
- Ji, Z., Yu, T., Xu, Y., Lee, N., Ishii, E., Fung, P., 2023. Towards mitigating LLM hallucination via self reflection. In: Findings of the Association for Computational Linguistics. EMNLP 2023, pp. 1827–1843.
- Jiang, H., He, P., Chen, W., Liu, X., Gao, J., Zhao, T., 2019. Smart: Robust and efficient fine-tuning for pre-trained natural language models through principled regularized optimization. arXiv preprint arXiv:1911.03437.
- Jiang, T., Huang, S., Luo, S., Zhang, Z., Huang, H., Wei, F., et al., 2024. MoRA: High-rank updating for parameter-efficient fine-tuning. arXiv preprint arXiv:2405.12130.

- Jiang, A.Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D.S., Casas, D.D.L., et al., 2023. Mistral 7B. arXiv preprint [arXiv:2310.06825](#).
- Jin, Q., Dhingra, B., Liu, Z., Cohen, W.W., Lu, X., 2019. Pubmedqa: A dataset for biomedical research question answering. arXiv preprint [arXiv:1909.06146](#).
- Jin, H., Wei, W., Wang, X., Zhang, W., Wu, Y., 2023. Rethinking learning rate tuning in the era of large language models. In: 2023 IEEE 5th International Conference on Cognitive Machine Intelligence. CogMI, IEEE, pp. 112–121.
- Kim, J., Lee, J.H., Kim, S., Park, J., Yoo, K.M., Kwon, S.J., Lee, D., 2024a. Memory-efficient fine-tuning of compressed large language models via sub-4-bit integer quantization. Adv. Neural Inf. Process. Syst. 36.
- Kim, S., Yang, H., Kim, Y., Hong, Y., Park, E., 2024b. Hydra: Multi-head low-rank adaptation for parameter efficient fine-tuning. Neural Netw. 106414.
- Koohpayegani, S.A., Navaneeth, K.L., Nooralinejad, P., Kolouri, S., Pirsiavash, H., 2024. NOLA: compressing LoRA using linear combination of random basis. In: The Twelfth International Conference on Learning Representations.
- Kwon, S.J., Kim, J., Bae, J., Yoo, K.M., Kim, J.H., Park, B., et al., 2022. Alphas-tuning: Quantization-aware parameter-efficient adaptation of large-scale pre-trained language models. arXiv preprint [arXiv:2210.03858](#).
- Li, M., Chen, L., Chen, J., He, S., Gu, J., Zhou, T., 2024a. Selective reflection-tuning: Student-selected data recycling for llm instruction-tuning. arXiv preprint [arXiv:2402.10110](#).
- Li, Z., Li, X., Liu, Y., Xie, H., Li, J., Wang, F.L., et al., 2023a. Label supervised llama finetuning. arXiv preprint [arXiv:2310.01208](#).
- Li, X.L., Liang, P., 2021. Prefix-tuning: Optimizing continuous prompts for generation. arXiv preprint [arXiv:2101.00190](#).
- Li, D., Ma, Y., Wang, N., Cheng, Z., Duan, L., Zuo, J., et al., 2024b. Mixlora: Enhancing large language models fine-tuning with lora based mixture of experts. arXiv preprint [arXiv:2404.15159](#).
- Li, J., Pan, K., Ge, Z., Gao, M., Ji, W., Zhang, W., et al., 2023b. Fine-tuning multimodal llms to follow zero-shot demonstrative instructions. In: The Twelfth International Conference on Learning Representations.
- Li, Q., Yang, X., Wang, H., Wang, Q., Liu, L., Wang, J., et al., 2023d. From beginner to expert: Modeling medical knowledge into general llms. arXiv preprint [arXiv:2312.01040](#).
- Li, Y., Yu, Y., Liang, C., He, P., Karampatziakis, N., Chen, W., Zhao, T., 2023c. Loftq: Lora-fine-tuning-aware quantization for large language models. arXiv preprint [arXiv:2310.08659](#).
- Lialin, V., Deshpande, V., Rumshisky, A., 2023. Scaling down to scale up: A guide to parameter-efficient fine-tuning. arXiv preprint [arXiv:2303.15647](#).
- Liao, B., Tan, S., Monz, C., 2024. Make pre-trained model reversible: From parameter to memory efficient fine-tuning. Adv. Neural Inf. Process. Syst. 36.
- Lin, C.Y., 2004. Rouge: A package for automatic evaluation of summaries. In: Text Summarization Branches Out. pp. 74–81.
- Liu, Y., 2019. Roberta: A robustly optimized bert pretraining approach. arXiv preprint [arXiv:1907.11692](#).
- Liu, Z., Kundu, S., Li, A., Wan, J., Jiang, L., Beerel, P.A., 2024a. AFLoRA: Adaptive freezing of low rank adaptation in parameter efficient fine-tuning of large models. arXiv preprint [arXiv:2403.13269](#).
- Liu, Z., Lyn, J., Zhu, W., Tian, X., Graham, Y., 2024c. Alora: Allocating low-rank adaptation for fine-tuning large language models. arXiv preprint [arXiv:2403.16187](#).
- Liu, W., Qiu, Z., Feng, Y., Xiu, Y., Xue, Y., Yu, L., et al., 2023. Parameter-efficient orthogonal finetuning via butterfly factorization. arXiv preprint [arXiv:2311.06243](#).
- Liu, Q., Wu, X., Zhao, X., Zhu, Y., Xu, D., Tian, F., Zheng, Y., 2024d. When MOE meets LLMs: Parameter efficient fine-tuning for multi-task medical applications. In: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 1104–1114.
- Liu, Y., Zhu, Z., Gong, C., Cheng, M., Hsieh, C.J., You, Y., 2024b. Sparse mezo: Less parameters for better performance in zeroth-order llm fine-tuning. arXiv preprint [arXiv:2402.15751](#).
- Luo, T., Lei, J., Lei, F., Liu, W., He, S., Zhao, J., Liu, K., 2024. Moelora: Contrastive learning guided mixture of experts on parameter-efficient fine-tuning for large language models. arXiv preprint [arXiv:2402.12851](#).
- Lv, K., Yang, Y., Liu, T., Gao, Q., Guo, Q., Qiu, X., 2023. Full parameter fine-tuning for large language models with limited resources. arXiv preprint [arXiv:2306.09782](#).
- Malladi, S., Gao, T., Nichani, E., Damian, A., Lee, J.D., Chen, D., Arora, S., 2023. Fine-tuning language models with just forward passes. Adv. Neural Inf. Process. Syst. 36, 53038–53075.
- Martino, A., Iannelli, M., Truong, C., 2023. Knowledge injection to counter large language model (LLM) hallucination. In: European Semantic Web Conference. Springer Nature Switzerland, Cham, pp. 182–185.
- Matthews, B.W., 1975. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. Biochim. Biophys. Acta (BBA)- Protein Struct. 405 (2), 442–451.
- Meng, X., Dai, D., Luo, W., Yang, Z., Wu, S., Wang, X., et al., 2024. Periodiclora: Breaking the low-rank bottleneck in lora optimization. arXiv preprint [arXiv:2402.16141](#).
- Mihaylov, T., Clark, P., Khot, T., Sabharwal, A., 2018. Can a suit of armor conduct electricity? A new dataset for open book question answering. arXiv preprint [arXiv:1809.02789](#).
- Nikdan, M., Tabesh, S., Alistarh, D., 2024. Rosa: Accurate parameter-efficient fine-tuning via robust adaptation. arXiv preprint [arXiv:2401.04679](#).
- Ning, L., Liu, L., Wu, J., Wu, N., Berlowitz, D., Prakash, S., et al., 2024. User-LLM: efficient LLM contextualization with user embeddings. arXiv preprint [arXiv:2402.13598](#).
- Novikova, J., Dušek, O., Rieser, V., 2017. The E2E dataset: New challenges for end-to-end generation. arXiv preprint [arXiv:1706.09254](#).
- Papineni, K., Roukos, S., Ward, T., Zhu, W.J., 2002. Bleu: a method for automatic evaluation of machine translation. In: Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics. pp. 311–318.
- Qiang, R., Zhang, R., Xie, P., 2024. BiLoRA: A bi-level optimization framework for overfitting-resilient low-rank adaptation of large pre-trained models. arXiv preprint [arXiv:2403.13037](#).
- Qin, H., Ma, X., Zheng, X., Li, X., Zhang, Y., Liu, S., et al., 2024. Accurate lora-finetuning quantization of llms via information retention. arXiv preprint [arXiv:2402.05445](#).
- Qiu, Z., Liu, W., Feng, H., Xue, Y., Feng, Y., Liu, Z., et al., 2023. Controlling text-to-image diffusion by orthogonal finetuning. Adv. Neural Inf. Process. Syst. 36, 79320–79362.
- Rajabzadeh, H., Valipour, M., Zhu, T., Tahaei, M., Kwon, H.J., Ghodsi, A., et al., 2024. Qdylora: Quantized dynamic low-rank adaptation for efficient large language model tuning. arXiv preprint [arXiv:2402.10462](#).
- Rajpurkar, P., 2016. Squad: 100,000+ questions for machine comprehension of text. arXiv preprint [arXiv:1606.05250](#).
- Ratner, N., Levine, Y., Belinkov, Y., Ram, O., Magar, I., Abend, O., et al., 2022. Parallel context windows for large language models. arXiv preprint [arXiv:2212.10947](#).
- Renduchintala, A., Konuk, T., Kuchaiev, O., 2023. Tied-lora: Enhancing parameter efficiency of lora with weight tying. arXiv preprint [arXiv:2311.09578](#).
- Shen, W., Li, C., Chen, H., Yan, M., Quan, X., Chen, H., et al., 2024. Small llms are weak tool learners: A multi-llm agent. arXiv preprint [arXiv:2401.07324](#).
- Shi, S., Huang, S., Song, M., Li, Z., Zhang, Z., Huang, H., et al., 2024. Reslora: Identity residual mapping in low-rank adaptation. arXiv preprint [arXiv:2402.18039](#).
- Shridhar, K., Sinha, K., Cohen, A., Wang, T., Yu, P., Pasunuru, R., et al., 2023. The art of llm refinement: Ask, refine, and trust. arXiv preprint [arXiv:2311.07961](#).
- Sinha, S., Yue, Y., Soto, V., Kulkarni, M., Lu, J., Zhang, A., 2024. MAML-en-LLM: Model agnostic meta-training of LLMs for improved in-context learning. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. pp. 2711–2720.
- Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C.D., Ng, A.Y., Potts, C., 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In: Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing. pp. 1631–1642.
- Somayajula, S.A., Liang, Y., Singh, A., Zhang, L., Xie, P., 2024. Generalizable and stable finetuning of pretrained language models on low-resource texts. arXiv preprint [arXiv:2403.12918](#).
- Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., Liu, Y., 2024. Roformer: Enhanced transformer with rotary position embedding. Neurocomputing 568, 127063.
- Sutskever, I., 2014. Sequence to sequence learning with neural networks. arXiv preprint [arXiv:1409.3215](#).
- Tan, T.F., Elangovan, K., Jin, L., Jie, Y., Yong, L., Lim, J., et al., 2024. Fine-tuning large language model (LLM) artificial intelligence chatbots in ophthalmology and LLM-based evaluation using GPT-4. arXiv preprint [arXiv:2402.10083](#).
- Tang, N., Fu, M., Zhu, K., Wu, J., 2024. Low-rank attention side-tuning for parameter-efficient fine-tuning. arXiv preprint [arXiv:2402.04009](#).
- Tayaranian, M., Ghaffari, A., Tahaei, M.S., Rezagholizadeh, M., Asgharian, M., Nia, V.P., 2022. Towards fine-tuning pre-trained language models with integer forward and backward propagation. arXiv preprint [arXiv:2209.09815](#).
- Team, G., Mesnard, T., Hardin, C., Dadashi, R., Bhupatiraju, S., Pathak, S., et al., 2024. Gemma: Open models based on gemini research and technology. arXiv preprint [arXiv:2403.08295](#).
- Tian, C., Shi, Z., Guo, Z., Li, L., Xu, C., 2024. HydraLoRA: an asymmetric LoRA architecture for efficient fine-tuning. arXiv preprint [arXiv:2404.19245](#).
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.A., Lacroix, T., et al., 2023b. Llama: Open and efficient foundation language models. arXiv preprint [arXiv:2302.13971](#).
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., et al., 2023a. Llama 2: Open foundation and fine-tuned chat models. arXiv preprint [arXiv:2307.09288](#).
- Valipour, M., Rezagholizadeh, M., Kobayev, I., Ghodsi, A., 2022. Dylora: Parameter efficient tuning of pre-trained models using dynamic search-free low-rank adaptation. arXiv preprint [arXiv:2210.07558](#).
- Vaswani, A., 2017. Attention is all you need. Adv. Neural Inf. Process. Syst.
- Vedantam, R., Zitnick, C.Lawrence., Parikh, D., 2015. Cider: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 4566–4575.
- Wang, Y., Agarwal, S., Mukherjee, S., Liu, X., Gao, J., Awadallah, A.H., Gao, J., 2022b. AdaMix: Mixture-of-adaptations for parameter-efficient model tuning. arXiv preprint [arXiv:2205.12410](#).
- Wang, B., Komatsuzaki, A., 2021. GPT-J-6B: A 6 billion parameter autoregressive language model.
- Wang, Y., Si, S., Li, D., Lukasik, M., Yu, F., Hsieh, C.J., et al., 2022a. Preserving in-context learning ability in large language model fine-tuning.

- Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., Bowman, S.R., 2018. GLUE: A multi-task benchmark and analysis platform for natural language understanding. arXiv preprint [arXiv:1804.07461](#).
- Wang, Y., Wu, J., Dabral, T., Zhang, J., Brown, G., Lu, C.T., et al., 2023. Non-intrusive adaptation: Input-centric parameter-efficient fine-tuning for versatile multimodal modeling. arXiv preprint [arXiv:2310.12100](#).
- Wang, C., Yang, Y., Li, R., Sun, D., Cai, R., Zhang, Y., Fu, C., 2024. Adapting llms for efficient context processing through soft prompt compression. In: Proceedings of the International Conference on Modeling, Natural Language Processing and Machine Learning. pp. 91–97.
- Warstadt, A., 2019. Neural network acceptability judgments. arXiv preprint [arXiv:1805.12471](#).
- Wei, J., Yao, Y., Ton, J.F., Guo, H., Estornell, A., Liu, Y., 2024. Measuring and reducing llm hallucination without gold-standard answers via expertise-weighting. arXiv preprint [arXiv:2402.10412](#).
- Wen, Z., Zhang, J., Fang, Y., 2024. SIBO: A simple booster for parameter-efficient fine-tuning. arXiv preprint [arXiv:2402.11896](#).
- Williams, A., Nangia, N., Bowman, S.R., 2017. A broad-coverage challenge corpus for sentence understanding through inference. arXiv preprint [arXiv:1704.05426](#).
- Wu, Z., Arora, A., Wang, Z., Geiger, A., Jurafsky, D., Manning, C.D., Potts, C., 2024a. Reft: Representation finetuning for language models. arXiv preprint [arXiv:2404.03592](#).
- Wu, S., Fei, H., Qu, L., Ji, W., Chua, T.S., 2023. Next-gpt: Any-to-any multimodal llm. arXiv preprint [arXiv:2309.05519](#).
- Wu, H., Zheng, H., Yu, B., 2024b. Parameter-efficient sparsity crafting from dense to mixture-of-experts for instruction tuning on general tasks. arXiv preprint [arXiv:2401.02731](#).
- Xia, W., Qin, C., Hazan, E., 2024. Chain of lora: Efficient fine-tuning of language models via residual learning. arXiv preprint [arXiv:2401.04151](#).
- Xu, R., Luo, F., Zhang, Z., Tan, C., Chang, B., Huang, S., Huang, F., 2021. Raise a child in large language model: Towards effective and generalizable fine-tuning. arXiv preprint [arXiv:2109.05687](#).
- Xu, Y., Xie, L., Gu, X., Chen, X., Chang, H., Zhang, H., et al., 2023b. Qa-lora: Quantization-aware low-rank adaptation of large language models. arXiv preprint [arXiv:2309.14717](#).
- Xu, L., Xie, H., Qin, S.Z.J., Tao, X., Wang, F.L., 2023a. Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment. arXiv preprint [arXiv:2312.12148](#).
- Yang, Y., Yuan, S., Cer, D., Kong, S.Y., Constant, N., Pilar, P., et al., 2018. Learning semantic textual similarity from conversations. arXiv preprint [arXiv:1804.07754](#).
- Ye, Z., Li, D., Tian, J., Lan, T., Zuo, J., Duan, L., et al., 2023. ASPEN: high-throughput LoRA fine-tuning of large language models with a single gpu. arXiv preprint [arXiv:2312.02515](#).
- Yuan, H., Yuan, Z., Tan, C., Huang, F., Huang, S., 2022. Hype: Better pre-trained language model fine-tuning with hidden representation perturbation. arXiv preprint [arXiv:2212.08853](#).
- Zadouri, T., Üstün, A., Ahmadian, A., Ermiş, B., Locatelli, A., Hooker, S., 2023. Pushing mixture of experts to the limit: Extremely parameter efficient moe for instruction tuning. arXiv preprint [arXiv:2309.05444](#).
- Zafir, O., Boudoukh, G., Izsak, P., Wasserblat, M., 2019. Q8bert: Quantized 8 bit bert. In: 2019 Fifth Workshop on Energy Efficient Machine Learning and Cognitive Computing-NeurIPS Edition. EMC2-NIPS, IEEE, pp. 36–39.
- Zaheer, M., Guruganesh, G., Dubey, K.A., Ainslie, J., Alberti, C., Ontanon, S., et al., 2020. Big bird: Transformers for longer sequences. Adv. Neural Inf. Process. Syst. 33, 17283–17297.
- Zaken, E.B., Ravfogel, S., Goldberg, Y., 2021. Bitfit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. arXiv preprint [arXiv:2106.10199](#).
- Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., Choi, Y., 2019. Hellaswag: Can a machine really finish your sentence? arXiv preprint [arXiv:1905.07830](#).
- Zhang, Q., Chen, M., Bukharin, A., Karampatziakis, N., He, P., Cheng, Y., et al., 2023a. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning. arXiv preprint [arXiv:2303.10512](#).
- Zhang, R., Han, J., Liu, C., Gao, P., Zhou, A., Hu, X., et al., 2023b. Llama-adaptor: Efficient fine-tuning of language models with zero-init attention. arXiv preprint [arXiv:2303.16199](#).
- Zhang, F., Li, L., Chen, J., Jiang, Z., Wang, B., Qian, Y., 2023c. Increlora: Incremental parameter allocation method for parameter-efficient fine-tuning. arXiv preprint [arXiv:2308.12043](#).
- Zhang, H., Li, G., Li, J., Zhang, Z., Zhu, Y., Jin, Z., 2022a. Fine-tuning pre-trained language models effectively by optimizing subnetworks adaptively. Adv. Neural Inf. Process. Syst. 35, 21442–21454.
- Zhang, F., Pilanci, M., 2024. Spectral adapter: fine-tuning in spectral space. arXiv preprint [arXiv:2405.13952](#).
- Zhang, R., Qiang, R., Somayajula, S.A., Xie, P., 2024b. AutoLoRA: automatically tuning matrix ranks in low-rank adaptation based on meta learning. arXiv preprint [arXiv:2403.09113](#).
- Zhang, S., Roller, S., Goyal, N., Artetxe, M., Chen, M., Chen, S., et al., 2022b. Opt: Open pre-trained transformer language models. arXiv preprint [arXiv:2205.01068](#).
- Zhang, Z., Yang, F., Jiang, Z., Chen, Z., Zhao, Z., Ma, C., et al., 2024c. Position-aware parameter efficient fine-tuning approach for reducing positional bias in LLMs. arXiv preprint [arXiv:2404.01430](#).
- Zhang, Z., Zhao, D., Miao, X., Oliaro, G., Li, Q., Jiang, Y., Jia, Z., 2024a. Quantized side tuning: Fast and memory-efficient tuning of quantized large language models. arXiv preprint [arXiv:2401.07159](#).
- Zheng, J., Hong, H., Wang, X., Su, J., Liang, Y., Wu, S., 2024. Fine-tuning large language models for domain-specific machine translation. arXiv preprint [arXiv:2402.15061](#).
- Zhu, F., Liu, Z., Feng, F., Wang, C., Li, M., Chua, T.S., 2024. Tat-llm: A specialized language model for discrete reasoning over tabular and textual data. arXiv preprint [arXiv:2401.13223](#).
- Zi, B., Qi, X., Wang, L., Wang, J., Wong, K.F., Zhang, L., 2023. Delta-lora: Fine-tuning high-rank parameters with the delta of low-rank matrices. arXiv preprint [arXiv:2309.02411](#).